

UNIT – V

PREVALENT PRACTICES OF ASSESSMENT & REPORTING OF QUANTITATIVE DATA

5:00 Introduction

We know that 'assessment of learning' refers to gathering information on the learning progress / proficiency achieved by students whereas 'evaluation of learning' refers to judging the level of learning achievement of students on the basis of the data obtained in assessment. Teaching for good learning is not possible without having an effective assessment practice. Hence assessment should be integrated with the teaching - learning process. An effective assessment system is the one which enhances the quality of both the teaching and learning. But in our present education system, the assessment practices, as criticized by many educationists, encourage students to memorize the learning content and recall them efficiently rather than giving importance for developing various abilities and skills.

In this unit, the Drawbacks of the present assessment system, Assessment for better learning, Assessment for confidence building in students and Assessment for creative learners besides explaining the process by which the collected data regarding the

learning achievement of students are subjected to quantitative analysis.

The results obtained by subjecting the data regarding students' learning and achievement to quantitative analysis are presented in the form of a report. The format of the report varies depending upon the person for whom it is meant. Based on such reports students, teachers, parents and administrators of the educational institution can decide on the future course of student learning.

The Concept of 'reflective journal', Structure of a reflective journal, Aspects to be included in reflective journal, Advantages and disadvantages of reflective journal, Student learning portfolio and using it to assess student learning, its advantages and disadvantages, Statistical techniques of interpreting and reporting the data particularly the measures of central tendency, dispersion and correlation, Graphical and diagrammatic representation of data etc. are also to be discussed in detail.

5:01 Present Practices of Assessment of Students' Learning

Following the enactment of 'Right to Education Act in 2009', 'Continuous and Comprehensive Evaluation' (CCE) system was introduced in our country, from the academic year 2011, in the secondary education from Std VI to IX. This led to the development of an evaluation system to assess the different aspects of

students' growth and development by assessing both the 'Scholastic Proficiency' and the 'Co-Scholastic Achievement' of students throughout the academic year, through formative as well as summative assessments of student-learning. This is school-based assessment system of student-learning. However, the traditional assessment system of written examinations still continue to exist for Std. X, XI and XII.

In the prevalent assessment system till 2017, for Std. XI annual examinations are held to assess students' learning achievement, at the school level and for Stds. X and XII public examinations are conducted at the State level by the Board of Secondary Education. However, from the academic year 2017-18, the Education Department of Tamilnadu Government has been conducting public examination at the end of Std. XI also.

5:02 Drawbacks of the Present Assessment System of School Education

5:02:1 Shortcomings Found in the CCE System Implemented

The CCE system that is presently followed till IX Std. has not been well received by teachers. The major reason for this, is they have not fully understood the CCE system. As per the survey conducted by 'The Hindu', a leading English daily, among the school teachers, 67% have not well understood about the CCE system and 58% oppose the system. The probable

reason for this, may be the introduction of CCE without adequate preparation. The pre-requisite for the success of the CCE system is the development of adequate awareness among the teachers about the CCE, imparting proper training and ensuring through evaluation whether teachers have understood the CCE.

Further, as findings of surveys indicate, the student - teacher ratio in most of the Indian schools is 42:1 as against the RTE stipulation of 30:1. At present there is a shortage of about 15 lakhs trained and fully qualified teachers. In primary schools, most of the teachers are engaged in duties other than teaching and assessing student learning. Hence, CCE has been given a raw deal by them and they just fill up the assessment forms for the name sake.

CCE, started with the aim of eliminating the competitive attitude as well as the examination stress among the students, has not achieved its purpose; instead it has increased the workload of teachers and made them to suffer from mental depression. Further, in the assessment of student learning, there is more subjectivity rather than being objective.

Due to the above cited reasons, the CCE has failed to achieve the purpose for which it was brought in. Research findings reveal that students studying in V Std. are not able to read fluently the III Std. text books.

Other than the above, the following are the reasons for CCE not able to achieve success as expected.

- i) Frequent assessment tests undertaken in the CCE system has increased the mental stress of students.
- ii) There is no uniformity in the evaluative procedures and they differ from school to school.
- iii) CCE is not suitable for over crowded classes with more number of students.
- iv) Large number of students are given 'A' grade (75% marks to 89%) representing high level of learning proficiency achieved, though most of them do not deserve it. For example, those who pass out with distinctions in the X Std public examinations, join the science stream in the XI Std and struggle to get the minimum pass mark in the XII Std. public examinations.
- v) Students, besides lacking subject knowledge, also lag behind in their language skills.
- vi) Teachers get engaged more in the clerical work of filling up the various assessment forms in large numbers, rather than in the preparation of regular classroom teaching. Teachers' complaint is that they do not get adequate time for the preparation of classroom teaching.

Thus whatever may be the cause, the harsh reality is that CCE has not contributed to increasing the standard of school education but it has led to its deterioration.

5:02:2 Drawbacks of the Written Examination System of School Education

CCE system is implemented for Std. VI to IX and the old written examination system still continues for students of Std. X, XI and XII. The following are the drawbacks of assessing students' learning achievement only through written examinations.

- i) Assessing students' learning achievement at the end of the year, without taking into consideration how well they involved themselves in learning, is highly objectionable, as some students who are regular in their studies may become nervous or fall sick on the eve of annual examination due to one reason or the other and unable to express their full potential in the written examinations. On the other hand, those who have not involved themselves in learning throughout the year, may have the luck to find questions in the examination, from few content areas which alone they have learned and score heavily by answering them well.
- ii) Examination questions too are such that, they encourage students to memorize the facts in the lesson, and recall efficiently; they do not test

students' higher order cognitive abilities like understanding, application, analysis, synthesis and evaluation.

- iii) Examinations test only the bookish knowledge of students and not their problem - solving ability or practical skills. Even those who successfully pass the higher secondary public examinations are not able to fill up properly the bank challans for money withdrawal or deposit.
- iv) Written examinations generally are such that, they do not encourage students' critical thinking and creativity.
- v) Assessing students' learning achievement through written examinations, lack objectivity and reliability.
- vi) Our examination system is such that students' subject knowledge is assessed through written examination for 80% of marks and practical examinations are held for the remaining 20% of marks.

Stated briefly, our assessment system of student learning, tests only the bookish knowledge and not the skills required to face their future life; further, it does not serve to enhance students' learning. The overall result is, students receive education that largely remains unrelated to their future life.

5:03 Assessment for Promoting Better Learning

Assessment of student learning is not an end in itself i.e. assessment does not stop with evaluating students' learning achievement and grading their performance. Based on the findings of assessment of student-learning, the teacher is required to provide students appropriate feedback enabling them to know the shortcomings in their learning and the means and methods of removing them. Therefore, teacher providing feedback to students, is considered as an integral part of assessment of student learning. Classroom teaching followed by assessment of student learning, teacher providing students appropriate feedback based on the findings of the assessment, students taking efforts to remove the shortcomings in their learning based on the feedback they received, assessing the learning achievement of students again, after they refined it, further classroom instruction etc. form the cyclic activities in the teaching- learning process.

Based on the assessment of student learning, those who are identified as significantly lagging behind others in their learning achievement, should be

subjected to a diagnostic test, conducted in the subject content taught and the teacher is required to take up remedial measures based on the findings obtained from the diagnostic test. Stated briefly, the feedback that follows the assessment is an integral part of assessment; teaching and assessment are integrated together. The end result of assessment of students is the enhancement of their learning.

Summative assessment held at the end of the instructional period such as the annual examinations are helpful only to evaluate students' learning achievement and grade them as well as monitor the standard of teaching in schools; they do not lend themselves much for enhancing students' learning.

Formative assessment that are undertaken by the teacher during the instructional period are alone highly useful in enhancing student-learning. The oral questions put forth by the teacher during his / her classroom teaching, assignments provided by the teacher daily at the terminal stage of his / her teaching, related to what are being taught that day, weekly achievement tests etc. are integrated with teaching; they alone serve as 'assessment for promoting better learning' in students. As quarterly examination and half yearly

examination too provide scope for giving feedback to students, after the assessment, they too can be considered helpful to improve student learning.

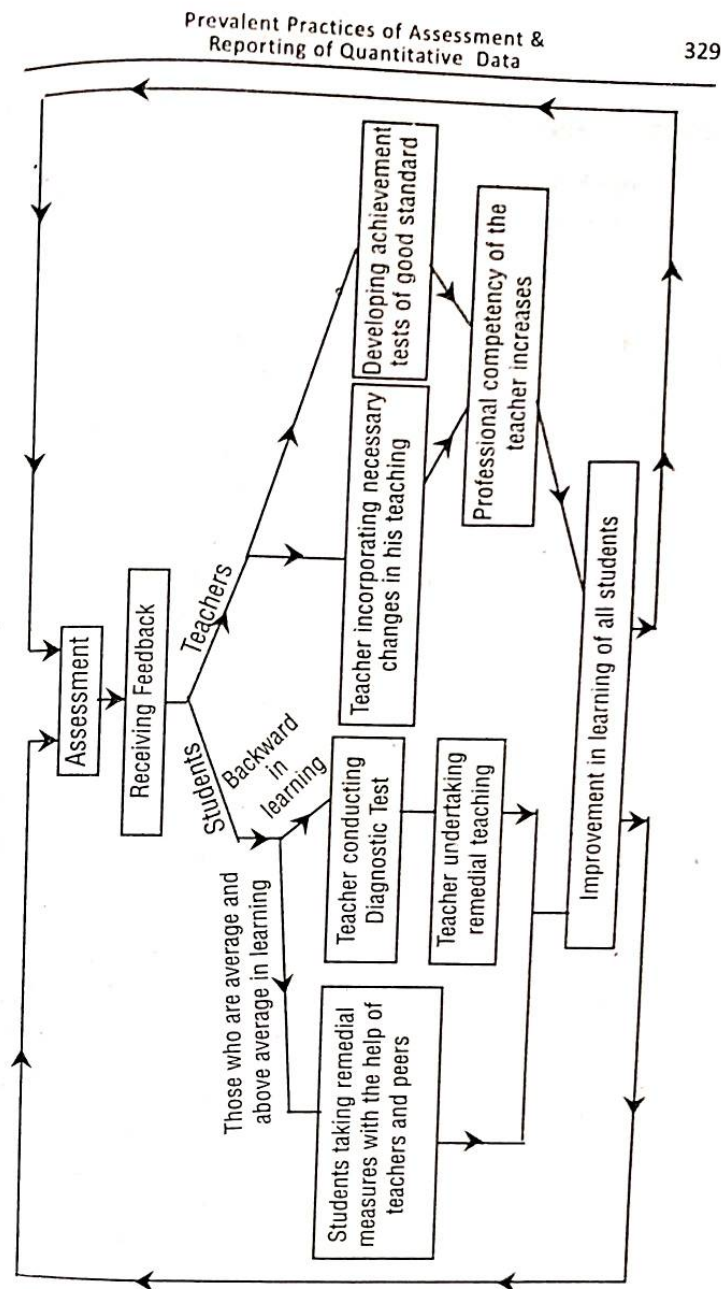
Assessment of student learning provides the teacher two kinds of feedback. When the teacher finds that most of the students in the class struggle to learn some facts / concepts in a lesson or particular content areas taught, he / she can undertake reteaching the same (by providing additional information and illustrations, making use of some more instructional resources and aids etc.) so as to improve student learning. If more of the students in the class fail to express high learning achievement in the assessment, then it implies, there are some shortcomings in the classroom teaching of the teacher. The teacher is required to incorporate necessary changes in his / her teaching so as to improve the quality of classroom instruction. This paves the way for the improvement in learning of all students in the class.

Suppose, if almost all students in the class express very high learning achievement, it implies the standard of the achievement test used in the evaluation of student learning is low. Most of the questions in the test may have been easy, testing only the cognitive

ability of 'knowing'. The feedback received from this makes the teachers understand the necessity of developing quality achievement tests with reasonable standard, by including suitable items which test higher order cognitive skills like 'understanding', 'application', 'reasoning' etc.

Summing up, it could be stated that the feedback provided, following the assessment of student learning, helps students to improve their learning by eliminating the shortcomings pointed out. Feedback is also useful for teachers in two ways. It helps the teacher to know the subject areas which students find it difficult to learn and take necessary remedial measures for the same. It also helps the teacher to know the deficiencies in his/her teaching so that the required changes could be adopted in the teaching so as to improve the quality of classroom instruction. The total effect of all these, results in the significant improvement in learning of all students in the class. Hence, assessment with appropriate feedback can be considered as a way to promote better learning in students.

The following diagram illustrates how assessment operates to promote better learning in students.



5:04 Assessment for Confidence Building

Those who have high self-confidence believe that they have the capacity to learn and do anything and everything and function accordingly. However good the classroom teaching may be, it may not yield fruitful results in students who lack self-confidence and the level of learning achievement of such students may not be high. Therefore, the most important professional duty of the teacher is to develop self-confidence in his / her students. For this, assessment of learning, particularly the formative assessment helps to a great extent. Formative assessment that is employed during the teaching of every unit in the subject, aims at improving student learning.

While introducing a topic at the beginning of the class, if the teacher asks simple questions connected to the topic, by relating them to the practical life, almost all students could answer them. This way of teacher putting simple questions to students and eliciting correct answers from them, develop self-confidence in the students. In the assignments that are given everyday at the end of classroom teaching, if the questions are arranged in such a way that, at the beginning few questions are simple and then the difficulty level of questions gradually increases, enabling all students to succeed in answering atleast some questions at the start so that the self-confidence gained in that drives them to take efforts to attempt answering the remaining difficult ones. Hence, in

examinations as well as assignments, when different kinds of questions are given, the teacher should see that they proceed from simple to complex, easy to difficult.

As the learning style of each student differs, the teacher should try to provide different kinds of questions, testing various learning skills such as answering in a page / paragraph, drawing and labelling / colouring the parts of a given thing, constructing models / learning aids, solving puzzles, finding the meaning of difficult words given using a dictionary, using the given words in the right context with an illustration etc. which provides a chance to every student to complete the assessment atleast partially. This instills self-confidence in the students.

Formative assessment enables students to know not only what they have learned but also how well they have learned them. Following the assessment the feedback that is provided by the teacher should not only point out the shortcomings in individual student's learning but also make students know the ways and means of removing them. Further, following the assessment, the teacher while providing feedback to each student should also point out the strengths and talents revealed in his / her learning achievement, which enables them to develop self-confidence. Students thus helped to gain self-confidence, will be motivated to take efforts in removing the deficiencies in their learning. This promotes better learning in students.

Putting in brief, if the assessment of students is such that it focuses on not only evaluating their learning achievement but also on revealing the efforts the students have put in to learn and the level of success achieved in their efforts and is followed by providing feedback which offers guidance to students how to increase the level of success in their learning efforts, then it helps to develop self-confidence in students and ultimately leading to the enhancement of their learning achievement.

The salient features of assessment that helps in developing students' self-confidence are :

- i) The tasks or test items that find place in the assessment of student learning should be so arranged in such a way that they proceed from simple to difficult ones.
- ii) Items chosen for the test that assess students' learning achievement should be such that, all students can answer atleast some of the questions in the test.
- iii) Employing assessment tools to suit the learning style of each student.
- iv) Following the assessment, providing suitable feedback to each student.
- v) While providing the feedback, not only taking into consideration the learning achievement of students but also focussing attention on appreciating the efforts taken for learning, besides

indicating the level of success achieved in their efforts so far and offering guidance in the ways and means to make their efforts more effective and fruitful.

- vi) All students in the class join together to develop the rubric for scoring the achievement test / assignment submitted and get the approval of the teacher.
- vii) Using the rubrics, all students in the class, evaluate their own answer scripts of the achievement test / assignment and also that of the peers.
- viii) Following the assessment of student learning, the teacher undertakes remedial teaching for those students who are found to be significantly backward in their learning and also arranges for peer tutoring with the help of those who have high proficiency in learning.

5:05 Assessment for Creative Learners

Among the students, some may be highly creative. They exhibit qualities like sensitivity to problems, ability to view any problem or issue from different angles, perceive remote associations or relationships even among apparently disconnected ideas and things, persistence, seeking to find new relationship among what have been learned, highly flexible in their thinking to cast ideas in different

patterns to produce new ideas or innovative solutions to problems, highly imaginative, raising questions about anything presented to them etc.

The assessment of learning of creative students will consist of activities like the teacher providing for suitable contexts in which creativity gets expressed and observing their behaviour, assessing their participation in group work, asking open-ended questions and analysing the replies given by them, making them participate in group discussions and analyse the ideas they express, making them to talk on a subject topic, at a short notice of five minutes, including a large number of test items in the assessment that test the cognitive skill of 'application', conducting quiz programmes related to the content areas of the subject, making them complete assignments and projects on suitable subject topics etc.

i) Assessing students' activities in Creative Contexts

Teacher can observe and assess students' activities in the following tasks which provide for creative contexts.

- a) In the language subject class, the teacher can present a topic and ask students to compose a poem on it, write a small essay on the given topic etc.
- b) In the mathematics class, the teacher can ask the students to solve the given puzzles; making

students to solve the given problems, particularly in arithmetic, using an alternative method rather than the one traditionally followed.

- c) In the science class the teacher can ask students to conduct experiments in the laboratory, testing the concepts taught in the class; ask them to prepare simple learning aids made from waste materials.

ii) Assessing through Group Work

Teacher can observe and assess students' activities in group work like sharing one's ideas with others, suggesting different possible solutions to a problem or an issue, approaching problems in alternate routes etc.

iii) Asking open-ended questions (providing for divergent thinking) and analysing the replies given

Instead of asking traditional questions (which mostly require convergent thinking) teacher can ask open-ended questions which provide opportunities for divergent thinking and analyse the replies given by students for their uniqueness.

(e.g.)

- a) Instead of asking the traditional question "What are the advantages of friction?", the teacher can shoot the open-ended question as follows : "What would happen if there is no friction at all in this world?"

- b) Instead of the traditional question "What are the benefits of the Himalayas to India?" the teacher can ask the following question : "What would have happened to India, if there is no Himalayan range of mountains?"
- c) Topics such as the following can be given to students to express their creative ideas
- If I become the Indian Prime Minister
 - If I become a scientist ...
 - If I get an opportunity to go to a different planet ...
- iv) **Providing a subject topic and asking students to talk on it at a short notice / conducting a quiz programme**

Only those students who have a good knowledge of the subject content and imaginative ability, can think over a given topic for a short while, select the points to be put forth, mentally integrate them in a logical order and talk on the topic fluently without any hesitation, before the teacher and peers in the class.

Similarly, quiz programmes could be conducted using the subject content. In this, only students with creative ability reply quickly to the questions asked.

- Teacher could observe and analyse the participation of creative students in different kinds of group work like 'group discussion', 'group project', 'debate' etc.
- Teacher should prepare achievement tests, giving more importance for 'application' type test items

rather than those which test students' ability to 'recall the facts from memory'.

Teachers could make use of such approaches and techniques, mentioned above to assess the learning of creative students in the class.

5:06 Reflective Journal

5:06:1 Concept of 'Reflective Journal'

A reflective journal is a means of recording ideas, personal thoughts and experiences, as well as reflections and insights a student have in the process of a course.

Reflective journal requires students to reflect every day, outside the classrooms in the evening or night, sitting alone at home, on the concepts they have learned in the classroom, the learning activities that are organized by the teacher, illustrations and explanations offered, discussions held with peers, reference books and materials suggested for further learning etc. and record the same in a separate note book, provided for each subject of study. Every student submit the notebook containing such notes prepared by him / her to the teacher at the end of each week. The teacher goes through the note books submitted to him / her, during the weekly holidays, records the comments regarding the strength and weaknesses of each student in their respective note books and returns the same, in the morning session of the first day of the following week, providing feedback either orally

or through the written comments recorded in the note books, about the ways and means of rectifying the shortcomings in their learning.

Students should be briefed at the beginning of the academic year itself that they should record briefly about all what they have learned in the class, such as the concepts and definitions for them, laws and principles, formulae and procedures, the way they have understood them, apply them in the practical life to know their utility, content areas which they felt difficult to understand, the activities / efforts needed that may make them understand those difficult content areas etc.

By going through students' reflective journal, the teacher can easily understand how learning has taken place in them i.e. reflective journal could be used as a means to assess student learning. It could be used both by the teacher and students as a method to improve learning.

5:06:2 Model Format of Reflective Journal

Name of the student :	Class and Section :
Subject :	Topic :
Date :	Class Period/Time :

1. Mentioning all the concepts that are learned in the class and the learning activities that are organized, related to those content taught in the class (i.e. mentioning the subject contents taught and the

learning experiences provided for them by the teacher).

2. Those that are easily understood among the contents that are taught.
3. Those that are felt difficult to understand among the contents that are taught.
4. Among the contents that are taught, those which they could relate with practical life.
5. Ways and means available to them to learn and understand those, which at present are difficult for them to learn.
6. Co-curricular activities related to learning, they are going to undertake.
7. If the concepts that are learned now are related to those already learned, then giving details about them.
8. Note regarding the doubts to be cleared, questions that are to be asked and specific problems that are to be pointed out if any.
9. Viewing totally, opinion about today's classroom learning.

Teacher's response / comments

Teacher's Signature

5:06:3 Illustration of a Reflective Journal

Name of the Pupil : Sureshbabu V	Class & Section : IX A
Subject : Physics	Topic : Newton's Laws of Motion
Date : 14.7.2017	Period : Morning II Period (10 - 10.45 a.m.)

S. No.	Topics in the lesson, Sub topics	Journal Entries
1.	Newtons first law of motion	i) Could understand the concept of 'inertia' of bodies implied in the Newton's first law of motion. ii) Difficult to understand the two kinds of inertia of bodies - (a) Static inertia and (b) Dynamic inertia or inertia at motion iii) Want to know the inertia of a body depends on what factors. iv) Could able to understand the concept of 'force' from the definition of inertia of a body; but feels further explanation is needed regarding the relationship between inertia of a body and force. v) Feels that Newton's I law of motion could be inferred as that one which states the definition of force.

S. No.	Topics in the lesson, Sub topics	Journal Entries
2.	Newton's second law of motion	i) Could understand that second law is defined as $F = m(v-u)/t$ or $F \propto ma$; however I could not understand properly while deriving the formula for force, why it is put as $F = Kma$ and how $F=ma$ comes from that. ii) I could work out the numerical problems presented by the teacher in the class, involving $F=ma$ and get the appreciation of the teacher. iii) I feel I have not fully understood the definition for the concept of 'force' regarding which I have to discuss with the teacher or other classmates. iv) I plan to get the reference books mentioned by the teacher, from the school library, in a day or two and go through them to get my doubts cleared.

5:06:4 Designing a Good Reflective Journal

1. Decide the format of reflective journal that fits the course contents in the subject (particularly being

student friendly) and explain about the same to students, at the beginning of the academic year itself.

2. Make sure that students understand the idea of a reflective journal i.e. making criteria on which their work will be assessed; the approximate length of each journal entry etc. i.e. make them realize the definition of 'reflection'.
3. Make sure students understand the purpose and benefits of reflective journals at the very beginning of the academic year.
4. Assure the students that the privacy and confidentiality of information provided by them will be safeguarded.
5. Make students understand the regularity of submission of their journal entries to the teacher, as decided at the beginning of the academic year.
6. Prepare rubrics for each topic in the subject, that finds place in the course of study.
7. Make sure feedback is provided to students based on the comments written in the reflective journal, while returning the same to them.

5:06:5 Rubrics for Assessing the Reflective Journal

Criteria for the award of assessment grades	Assessment Grades to be given			
	Very Good	Good	Average	Poor
Reflection Ability to integrate learning into real world experience and analyse issues with critical attitude.	i) Ability to proficiently demonstrate reflection and deep thinking of acquired knowledge and concepts and integrate them with different issues in the real life. ii) Demonstrating in their journal entries critical thinking skills; and thinking in new perspectives to get creative solution.	i) Showing satisfactory ability to relate acquired knowledge to previous experiences. ii) Demonstrating the ability to analyse the issues from a number of different perspectives.	i) Includes description of events and a little further consideration behind the events, using a relatively descriptive style of language. ii) No evidence of using multiple perspectives in analyzing the issues.	i) Includes only more descriptions of theoretical knowledge; no reflection is demonstrated beyond the descriptions.

Assessment Grades to be given				
Criteria for the award of assessment grades	Very Good	Good	Average	Poor
Presentation Articulation and organisation of the ideas and perspectives	i) Writing is well focussed; ii) Arguments or perspectives are precisely defined and explained. iii) Coherent flow in developing an insightful idea demonstrated.	i) Arguments or perspectives are clearly stated. ii) Organized flow in writing but not deep enough to be very insightful.	i) Arguments or perspectives are vaguely mentioned. ii) Writing lacked an organized flow and the ideas were hard to flow.	i) Do not show any original thinking or perspectives. ii) Chaotic in organization and presentation of ideas.
Completeness Incorporation of journal entries into a whole demonstration of the learning process.	Concrete connections between journal entries into a whole, demonstrating clear steps in the developmental process.	Journal entries are generally connected; still not able to observe how the student develops during the learning process.	i) Weak connections between journal entries; ii) development gained from the learning process is hardly observed.	i) No connections observed between journal entries. ii) The entries are mere description of events rather than showing a sequence of learning steps.

5:06:6 Advantages of Reflective Journal

1. Active Learning

The process of reflection encourages students to take the initiative to be active, self-driven and allows individuals learn to analyze issues / problems from different perspectives i.e. Students become independent thinkers through the practice of reflection, enabling them to solve various problems on their own.

2. Understanding the Progress of Students

Reflective journals provide good opportunities for teachers to understand about how students think and feel about their learning as well as the learning progress of students which results in the improvement of the classroom teaching process throughout the academic year.

3. Improving Writing Skills of Students

Writing reflective journals can involve students in a new form of writing which they may not have a chance to experience in the past. This exposure can bring out improvement in students' writing skills. Further reflective journals help teachers to assess the writing skills of students which also contribute in improving the writing skills of students.

4. Opportunities for Freely Expressing Personal Views and Gentle Criticisms

Reflective journals offer opportunities to students to express their personal views and criticisms about

the teaching-learning process that has taken place in the classroom. Generally students do not come forward to express themselves like this in the classroom and tutorials.

5. Enhancing Critical thinking and Creativity.

The process of self-reflection enhances the development of critical thinking skills among students when they relate their knowledge to real world issues. It also helps to develop their creativity and a questioning attitude towards different issues and problems.

5:06:7 Disadvantages of Reflective Journal

i) Difficult for objective marking

Due to subjective nature of reflective assignments, it is rather difficult for assessors to be objective and consistent in their grading. Different assessors when marking may have quite a large discrepancy in their judgement of different types of work.

ii) Time consuming for Grading

The context of reflective writing can often be very wide and involve a wide range of concepts, issues and perspectives. As a result, it often takes considerable amount of time for assessors to read and grade students' work.

iii) Confidentiality

As students have to disclose their personal and private views on a wide range of concepts, issues and

perspectives, some of them may be unwilling to honestly disclose their ideas, fearing it may affect the grade they receive.

iv) Clear guidelines needed

Many students may not be familiar with the procedure of writing reflective assignments. Teachers have to give clear guidelines to students about what should be included in the reflective journals, what can be learnt from writing it, as well as how they will be graded. How far these will be understood by students is a big question mark. How far it is possible to assess true learning progress of students using student reflective journal is also debatable.

5:07 Students' Learning Portfolio

Detailed description about this given in 2:02:4:02.

5:07:1 Assessment of Learning Using Student Portfolio

Detailed description about this given in 2:02:4:03.

5:07:2 Designing the Assessment of Learning Using Student Portfolio

Detailed description about this given in 2:02:4:04.

5:07:3 Advantages of Using Student Portfolio in the Assessment of Learning

This can be found in 2:02:4:05.

5:7:04 Disadvantages of Using Student Portfolio in the Assessment of Learning

This is explained in 2:02:4:06.

5:08 Analyzing, Interpreting and Reporting Quantitative Data

The assessment data concerning the learning achievement of students, obtained in the form of numerical measures, can not be used as such for interpreting the performance of students. For example, if a student gets an achievement score of 65 in Science, we can not say with certain, whether it is a good learning achievement or moderate / low learning achievement. We can interpret the achievement of that student only by knowing the measures of 'average' and 'dispersion' for the science scores of all those in the class who took part in the assessment test. If the 'average' mark for the whole class is 75, then the student's achievement score of 65 is not indicating satisfactory performance as compared to that of most others in the class. However, if the class average is 45, then the student's score of 65 is high compared to most others in the class and hence his/her performance is interpreted as 'Good'. Therefore, it becomes clear that for interpreting students' achievement scores, statistical analysis is required. Among the measures that find place in basic statistical analysis, the following are the important ones.

- i) Measures of Central tendency
- ii) Measures of Dispersion / Deviation
- iii) Correlation

In understanding basic or descriptive statistical analysis, the first task is to convert the individual achievement scores called '**raw data**' into a '**frequency distribution**', which consists of orderly arranged classes and class frequencies. Frequency distribution is usually presented in the form of a table called '**Frequency Distribution Table**'. Generally, data can be presented in the following three ways.

1. Textual Form
2. Tabular Form
3. Graphical Form

Among the above, data are mostly presented in the form of tables and graphs / diagrams.

Based on the findings obtained from statistical analysis, proper interpretation could be given for the students' achievement scores. Students' achievement scores in different school subjects, with appropriate interpretation is presented to students and their parents along with the feedback provided by the teacher.

Let us see in detail, the method of obtaining the three kinds of measures viz. Measures of Central tendency, Dispersion and Correlation, using basic statistical analysis of data.

5:09 Measures of Central Tendency

After arranging the individual data obtained, through the achievement test regarding students' learning achievement, in the form of classes and class frequencies (i.e. in the form of a frequency distribution) appropriate measures of central tendency are to be computed.

That single measure which reflects the nature of the entire sample of scores or the whole frequency distribution is called "Measure of Central Tendency".

It is the score around which scores are crowded. It is the representative of the given group of data. Just as we elect our representatives to reflect our voices in legislatures, measures of central tendency represent all the items in a given sample.

There are three "averages" or measures of central tendency that are in common use. They are :

- i) Arithmetic Mean
- ii) Median
- iii) Mode

5:09:1 Arithmetic Mean

The Arithmetic Mean or simply the Mean is the centre of gravity of the group. It is the fulcrum about which the entire group revolves. It is denoted by the letter M or $\bar{X}$. It is obtained by adding the values of all

items in the group and the resulting sum is divided by the number of items in the group. For example if the scores of 5 students are 60, 70, 80, 90 and 100 respectively, then the Mean of the group is

$$\bar{X} = \frac{60+70+80+90+100}{5} = \frac{400}{5} = 80$$

Formula for Mean (Raw Data)

If $X_1, X_2, X_3, X_4, \dots, X_n$ are the scores of n students in a given test, then the Mean of the group in the test is

$$M \text{ or } \bar{X} = \frac{X_1 + X_2 + X_3 + X_4 + \dots + X_n}{N}$$

$$\text{i.e., } \bar{X} = \frac{\Sigma X}{N}$$

Where ΣX denotes the "sum of the values" taken by the variable X and N is the total number of students.

5:09:1:01 Mean for the Discrete Series

A) Direct Method:

$$\text{Formula } \bar{X} = \frac{\Sigma fX}{N} \text{ where}$$

f = Frequency; $N = \Sigma f$ is the Values of the variable

Illustration

Compute mean for the following data :

Scores : 20, 25, 30, 35, 40

No. of students

who received

the above scores : 6, 10, 14, 6, 4

Marks obtained (X)	No. of students who received the score (f)	fX
20	6	120
25	10	250
30	14	420
35	6	210
40	4	160
	$N = \Sigma f = 40$	$\Sigma fX = 1160$

$$\bar{X} = \frac{\Sigma fX}{N} = \frac{1160}{40} = 29$$

B) Short-cut Method

Formula $\bar{X} = A + \frac{\Sigma fd}{N}$ Where A = Assumed Mean and d = X - A

Scores (X)	No. of students who received the score (f)	d = X - A (A = 30)	fd
20	6	-10	-60
25	10	-5	-50
<u>30</u> (A)	14	0	0
35	6	+5	+30
40	4	+10	+40
	N=40		$\Sigma fd = 70 - 110 = -40$

$$\bar{X} = 30 + \left(\frac{-40}{40} \right) = 30 + (-1) = 30 - 1 = 29$$

[Note: Choose any score (Preferably against the highest frequency) as assumed Mean A; then find the deviation d for each and every score given, by subtracting the value of A]

5:09:1:02 Mean for Grouped Data (Frequency Distribution)

(A) Direct Method (Long Method)

Formula $\bar{X} = \frac{\Sigma fX}{N}$ Where X refers to the value of the Mid-Point of Class Interval (C.I.) and N = Sum of the frequencies.

Illustration

Class Interval C.I.	Frequency (f)	Mid-point of C.I.	fX
1-10	3	5.5	16.5
11-20	5	15.5	77.5
21-30	6	25.5	153.0
31-40	4	35.5	142.0
41-50	6	45.5	273.0
51-60	12	<u>55.5</u> A	666.0
61-70	6	65.5	393.0
71-80	4	75.5	302.0
81-90	3	85.5	256.5
91-100	1	95.5	95.5
	$N = \Sigma f = 50$		$\Sigma fX = 2375$

$$\bar{X} = \frac{2375}{50} = 47.5$$

Procedure :

- Find Mid-point of each class interval as half of the sum of the **lower limit of the class + upper limit of the class**

- ii) Find the product of f and X for each class (i.e. fX)
- iii) Add the values fX and put the sum as ΣfX
- iv) Divide the sum of fX by N (i.e. $\frac{\Sigma fX}{N}$) to get the mean

B) Short-Cut Method

In this method, the mid-value of any one of the class intervals is taken as the assumed mean (A). Usually, to make computations easier, it is always located at the middle or the one against the highest frequency. For example, in the following illustration given, the highest frequency noticed is 12. Hence, the mid-point of the class interval 51-60 i.e. 55.5 is taken as the assumed mean (A).

Then, the step deviation ' d ' is calculated using the formula $d = \frac{X - A}{i}$ i.e. mid point of each class interval is how many units away from the assumed mean [1 unit = size of the class interval (i)] either increasing or decreasing. This is recorded in the fourth column of the table.

For computing it easily we put '0' against the assumed mean and put +1, +2, +3 ... etc. for the mid-values of class intervals which go on increasing; put -1, -2, -3 ... etc. for the mid-values of class intervals which go on decreasing. In the last column of the table, we record the product of ' f ' and ' d ' for the respective class interval. Then the sum of ' fd ' is computed and using that in the formula given, the mean for the given frequency distribution is to be calculated.

Illustration: Find the mean for the following distribution of scores.

Marks	1-10	11-20	21-30	31-40	41-50	51-60	61-70	71-80	81-90	91-100
No. of Students	3	5	6	4	6	12	6	4	3	1

Class Interval (C.I)	Frequency (f)	Mid-point of C.I (X)	$d = \frac{X - A}{i}$ (Step Deviation)	fd
1-10	3	5.5	-5	-15
11-20	5	15.5	-4	-20
21-30	6	25.5	-3	-18
31-40	4	35.5	-2	-8
41-50	6	45.5	-1	-6
51-60	12	(55.5) A	0	0
61-70	6	65.5	1	6
71-80	4	75.5	2	8
81-90	3	85.5	3	9
91-100	1	95.5	4	4
$i = 10$	$N = 50$			$\Sigma fd = 27 - 67 = -40$

$$\bar{X} = A + \left[\frac{\Sigma fd}{N} \times i \right] = 55.5 + \left[\frac{-40}{50} \times 10 \right] = 55.5 - 8 = 47.50$$

5:09:1:03 Merits of Arithmetic Mean

1. It has a well defined mathematical formula.
2. It depends on all the items in the distribution.
3. If we find the deviations of all the items in the given group from the 'mean' and add, its value is always zero. i.e. Mean serves as the centre of gravity of the given group of items. Hence 'mean' is considered as the measure of central tendency having greatest stability.
4. For the computation of other descriptive statistics like 'Standard deviation', 'coefficient of correlation' etc., mean is highly useful.

5:09:1:04 Demerits of Arithmetic Mean

1. Extreme items in a group of scores i.e. those having 'very small' or 'very high' values affect the 'mean' very much.
2. Usually, mean is not represented by the actual value of any of the items in the distribution.
3. If the exact value of any of the item in the distribution is unknown or if any item is missing in the distribution of scores then, mean cannot be computed, i.e. for incomplete frequency distribution mean cannot be calculated.

5:09:2 Median

Median is defined as the midpoint of a given distribution. It is the point which divides the given distribution exactly into two halves.

50%	Median
50%	

i.e., it is the point in a given distribution below which 50% of the total number of items lie.

5:09:2:01 Computation of Median for Raw Scores (Ungrouped Data)

Procedure :

- i) Write the given items in the ascending or descending order of values.
- ii) Median value = Value of the $\left(\frac{N+1}{2}\right)^{\text{th}}$ item, where
N = No. of total items given

Illustration - I (Where there are odd number of total items)

Compute the median for the given data :

96, 13, 74, 22, 65, 53, 49

Writing the items in the ascending order of values

13, 22, 49, 53, 65, 74, 96

Here N = 7

Value of the Median = Value of $\left(\frac{7+1}{2}\right)^{\text{th}}$ item
= Value of 4th item = 53

Illustration - II (Where there are even number of total items)

Given Data : 96, 13, 74, 22, 65, 53, 49, 37

Writing the items in the ascending order :

13, 22, 37, 49, 53, 65, 74, 96

Here $N = 8$

Value of the Median = Value of $\left[\frac{N+1}{2}\right]^{\text{th}}$ item
 = Value of $\left[\frac{8+1}{2}\right]^{\text{th}}$ item

$\Rightarrow$ Value of the 4.5th item

= Value of the 4th item + .5 (Difference between the 5th and 4th items)

$$= 49 + .5 (53 - 49) = 49 + .5 (4) = 49 + 2 = 51$$

5:09:2:02 Computation of Median for Grouped Data (Discrete Series)

Illustration

Compute Median for the following data :

Marks obtained :	20	25	30	35	40
No. of Students :	6	10	14	6	4

Marks (X)	No. of Students (Frequency) (f)	Cumulative Frequency (Cf)
20	6	6
25	10	16
30	14	30
35	6	36
40	4	40
	$N = \Sigma f = 40$	

$$\text{Median} = \left[\frac{N+1}{2}\right]^{\text{th}} \text{ item's value}$$

$$= \left[\frac{40+1}{2}\right]^{\text{th}} \text{ item's value}$$

$$= 20.5^{\text{th}} \text{ item's value}$$

= 30 (since the value of item number starting from 17 to 30, is 30).

5:09:2:03 Computation of Median for Grouped Data (Continuous Series or Frequency Distribution)

Procedure :

- Using given frequencies, starting from the lowest class interval, we should find the "less than cumulative frequency 'Cf'.
- Find where the value of $\left[\frac{N}{2}\right]^{\text{th}}$ item lies in cumulative frequency column. That C.I is called the median class. Mark the Median Class.
- Then use the formula and compute the median

$$\text{Mdn} = l + \left[\frac{\frac{N}{2} - C}{f} \times i \right]$$

Here l = Exact lower limit of the median class

C = Cumulative frequency upto the median class

i = Class size

f = Cumulative frequency of the median class

Illustration

Computation of Median for the frequency table given in illustration under 10:4:01:2

Class Interval (CI)	Frequency (f)	Cumulative Frequency (Cf)
1-10	3	3
11-20	5	8
21-30	6	14
31-40	4	18
41-50	6	24
51-60	12	36
61-70	6	42
71-80	4	46
81-90	3	49
91-100	1	50
$i=10$		

Here $\frac{N}{2} = \frac{50}{2} = 25$ which falls in the class interval 51-60, after the cumulative frequency 24. Therefore 51-60 is the 'median class' whose lower limit is 51 and the upper limit of the previous C.I is 50. Hence the true lower limit (l) of the C.I 51-60 is $\frac{51+50}{2} = 50.5$. The cumulative frequency before the C.I 51-60 is 24 and the frequency of the median class is 12.

i.e. $l = 50.5$; $C = 24$; $f = 12$; $N/2=25$; $i = 10$

Substituting these values in the formula

$$\begin{aligned}
 \text{Mdn} &= l + \left[\frac{\frac{N}{2} - C}{f} \times i \right] \\
 &= 50.5 + \left[\frac{25 - 24}{12} \times 10 \right] \\
 &= 50.5 + \left[\frac{1 \times 10}{12} \right] = 50.5 + .833 \\
 &= 51.33
 \end{aligned}$$

5:09:2:04 Merits of 'Median'

1. It is easy to calculate
2. It has a definite algebraic formula
3. It does not get affected by the small and big values of extreme items in a distribution
4. It can be calculated for open ended distributions too.
5. Even if the extreme items are missing or their values are not known exactly, even then median can be computed.
6. Median can also be calculated by graphical method.
7. In skewed distributions, 'median' is the more representative measure of central tendency than 'mean' and 'mode'.

5:09:2:05 Demerits of 'Median'

1. Median does not depend upon the values of all the items in the distribution.
2. It does not lend itself for further algebraic treatment.

- Median gets affected much by the variations caused by sampling of the items.
- It is not much useful for further statistical analysis.

5:09:3 Mode

Mode is that measure which occurs most frequently in a distribution. It is the popular measure or the most often repeated score in a distribution. Consider the series 25, 28, 32, 22, 20, 28, 30, 27, 30, and 28. In this group of scores, 28 gets repeated maximum number of times i.e. three times. Hence, the mode of this group of data is 28.

Consider another series 42, 48, 52, 37, 46, 42, 50, 37, 38. In this group of data, scores 42 and 37 get repeated two times which is the maximum number of times of occurrence. Hence both these scores are to be considered as 'modes'. Such type of distribution of scores are called 'Bimodal'.

Note : Don't take the average for the values of the modes in the case of multi-modal distributions.

5:09:3:01 Computation of Mode for Grouped Data (Discrete Series)

Illustration :

Find the mode for the following data :

Scores	:	20	25	30	35	40
No. of students	:	6	10	14	6	4

Solution

As the score 30 has the maximum value of frequency (i.e. 14), it is taken as the mode.

5:09:3:02 Computation of Mode for Grouped Data (Continuous Series)

Let us take the frequency distribution given under 5:09:1:02 as an example.

Solution :

Class Interval (CI)	Frequency (f)
1-10	3
11-20	5
21-30	6
31-40	4
41-50	6 $\leftarrow f_1$
Modal Class 51-60	12
61-70	6 $\leftarrow f_2$
71-80	4
81-90	3
91-100	1

$$N = \Sigma f = 50$$

Procedure :

- Class interval containing the maximum frequency is considered as the modal class. Here in our example, it is 51-60.
- The mid-point of the modal class is the approximate value of the mode. Here it is $\frac{51+60}{2} = 55.5$.
- The formula for computing Mode is $Me = l + \frac{if_2}{f_1 + f_2}$
Here l = True lower limit of the modal class

i = the class size; f_1 and f_2 are frequencies of the previous and succeeding class intervals of the modal class.

In our example, $l = 50.5$; $i = 10$; $f_1 = 6$, $f_2 = 6$

$$\begin{aligned} Me &= 50.5 + \frac{10 \times 6}{6 + 6} \Rightarrow 50.5 + \frac{60}{12} \\ &= 50.5 + 5 = 55.5 \end{aligned}$$

In a distribution, if there are two class intervals having equal value of maximum frequency, such a distribution is called '**Bimodal distribution**'. Distributions having more than two groups (C.Is) having equal value of maximum frequency, such distributions are known as '**multimodal distributions**'. The formula $Me = l + \frac{2f_2}{f_1 + f_2}$ can be employed to calculate 'mode', only for **mono-modal** distributions (i.e, distributions having only one mode).

For distributions having more than one mode, the following formula is used to compute the mode.

$$Me = 3 \text{ Median} - 2 \text{ Mean}$$

Illustration : If the mean and median of distribution are 50 and 52, then its mode = $3(52) - 2(50) = 56$.

Note :

- i) The distribution in which $M = Mdn = Me$, is called 'Normal Probability Distribution' i.e. most of the students in such a class have average achievement scores.

- ii) If $M < Mdn < Me$ then, such distributions are called negatively skewed i.e. most of the students in such a class are high achievers.
- iii) If $M > Mdn > Me$, then such distributions are known as positively skewed distributions i.e. most of the students in such a class are low achievers.

[Symbol ' $<$ ' denotes 'less than' and ' $>$ ' denotes 'greater than']

5:09:3:03 Merits of Mode

1. It is easy to understand and calculate.
2. The value of mode will not change due to very small or very large values in a distribution.
3. Mode can be found out using the graphical method.

5:09:3:04 Demerits of Mode

'Mode' does not find much use in educational research; in the business world it is used to find the size of commodity, which required to be manufactured in large quantities and the size of things which sellers can stock in large quantities. For example, chapels of size No. 9 and 10, shirts of size 40" and 42", banians of size 90 c.m. and 95 c.m. are in great demand in the consumer market and as such they are purchased and stocked by sellers in large quantities. Chapels of size number 10 or more and banians of size 100 c.m. and above are purchased and stocked by sellers in very small quantities.

5:10 Measures of Dispersion or Deviation

5:10:1 Concept of 'Measure of dispersion' and its Types

"Measures of dispersion (also known as measures of variability) are measures that describe the 'spread' or the 'scatter' of the individual scores in a group of data around an average or measure of central tendency".

Lesser the dispersion (or deviation), greater is the '**Homogeneity**' among the scores in the group and higher the dispersion, more is the "**Heterogeneity**" of the scores.

The following are the measures of 'Dispersion' used in analysis :

i) Range, (ii) Mean Deviation (M.D.), (iii) Standard Deviation (S.D.), and (iv) Quartile Deviation (Q.D.)

5:10:2 Need for the Measures of Dispersion

Averages alone are not sufficient to compare individuals or groups to know their complete nature. For example, let us consider the scores in the following two groups, each containing 10 students.

Group A : 15,25,35,45,55,55,65,75,85,95

Group B : 51, 52,53,54,55,55,56,57,58,59

Sum of the scores in both the groups are 550 and hence the means of the two groups are 50; the value of the median in both the groups are $\frac{55 + 55}{2} = 55$; the

mode is $3(55) - 2(55) = 55$. Thus all the three measures of 'average' are the same. Now the question arises, whether can we conclude that the proficiency level of both the groups are identical. The answer is, that we can not infer like that, since if we decide 40 is the minimum mark required for a pass in the examination, then in group 'A' three students get 'failed'; whereas in group 'B' all the 10 students pass the examination. If we think of awarding prizes for those who get 60 marks and above, then in group 'A' 4 students will get prizes and in group 'B' none will get a prize. The individual scores in group 'A' get deviated from the 'mean' 55 to a maximum of 40 units which is higher than the maximum deviation of scores in 'B' which is 4 units only.

Therefore it becomes clear that it is not proper to compare two groups, only on the basis of their respective 'mean'; along with the measures of average, measures of dispersion also should be taken into account while comparing scores or the performance of individuals or groups.

5:10:3 Range

Range is the difference between the value of the largest item (L) and the small item (S) in the group.

$$\text{Range } R = L - S$$

For example in the given two group of items

Group A : 15,25,35,45,55,55,65,75,85,95

Group B : 51, 52,53,54,55,55,56,57,58,59

Range of group 'A' is $95 - 15 = 80$

Range of group 'B' is $59 - 51 = 8$

We can easily understand the range for group 'A' is 10 times that of group 'B'

Merits :

It can be easily computed and understood.

Demerits :

- It is a true measure and is easily affected by fluctuations of sampling.
- It is not based on each and every item of the distribution.
- In the case of open-ended class distribution, range can not be computed.

5:10:4 Mean Deviation (M.D)

The mean deviation, also sometimes referred as **Average Deviation (A.D)** is the average of the absolute values of the deviations of individual items from their Arithmetic Mean. This is denoted by M.D.

The formula for computing M.D. for raw data is $M.D = \frac{\sum |x|}{N}$; Here $x = X - M$ (X is the value of each of the items, M is the mean and N is the total number of items in the group). For grouped data, the formula is $M.D = \frac{\sum f|x|}{N}$

[Note : If we put a number in between a set of parallel lines, it is known as 'Modules of the number, denoting its absolute value, devoid of any sign. For example $|+6|=6$; $|-6|=6$]

5:10:4:01 Computation of Mean Deviation of Ungrouped Data

Illustration :

Find the Mean deviation for the following:

$X : 12, 14, 16, 18, 20$

- In this, the mean of all the given scores is $80/5=16$

- In the second column, the deviation ' x ' of each and every item from the mean M (i.e. $x = X - M$) is computed.

Among the values obtained some are +ve and some are -ve.

- The sum of all the deviations i.e. $\sum x$ is zero.

- In the third column, the absolute value of each deviation i.e. $|x|$ is found out and recorded.

- Then the sum of $|x|$ is to be found out and divided by N .

In our example $M.D. = 12/5 = 2.4$

5:10:4:02 Computation of M.D for Grouped Data (Discrete Series)

Illustration

Find the Mean Deviation for the following data :

Marks (X) : 20 25 30 35 40

No. of Students (f) : 4 10 14 6 6

X	$x = X - M$	$ x $
12	-4	4
14	-2	2
16	0	0
18	2	2
20	4	4
$M = \frac{80}{5} = 16$	$\sum x = 6 - 6 = 0$	$\sum x = 12$

Solution

Marks (X)	No. of Students (f)	fX	x = X - M	x	f x
20	4	80	-10	10	40
25	10	250	-5	5	50
30	14	420	0	0	0
35	6	210	+5	5	30
40	6	240	+10	10	60
N = $\sum f = 40$		$\sum fX = 1200$	$\sum x = 0$		$\sum f x = 180$

$$\text{Mean} = M = \frac{\sum fX}{N} = \frac{1200}{40} = 30$$

$$\text{M.D} = \frac{\sum f|x|}{N} = \frac{180}{40} = 4.5$$

Procedure

- First find the Mean M by using the formula $M = \frac{\sum fX}{N}$
- In the fourth column of the table, the deviation of each item from the mean M ($x = X - M$) is computed and recorded. Among them some may be positive and some may be negative.
- In the fifth column, the positive and negative signs of each deviation value is removed and recorded.
- Each frequency is multiplied by the absolute deviation $|x|$ against it and the product $f|x|$ is recorded in the sixth column of the table.
- Then, sum of $f|x|$ is found out.
- Lastly, using the formula $\text{M.D.} = \frac{\sum f|x|}{N}$, mean deviation is computed.

5:10:4:03 Computation of M.D for Grouped Data (Continuous Series)

Illustration :

Find the Mean Deviation for the following data :

Marks : 1-10 11-20 21-30 31-40 41-50 51-60 61-70 71-80 81-90 91-100

No. of Students : 3 5 6 4 6 12 6 4 3 1

Steps in the Procedure :

- First find the mean using the formula

$$M = \frac{\sum fX}{N} ; \text{ in our example } M = \frac{2375}{50} = 47.5$$

- In the fifth

column, the deviation of mid-point of each C.I. from the mean ($x = X - M$) is computed and recorded.

- In the sixth

column, the absolute values of deviations (Numbers

devoid of signs) are to be recorded.

Class Interval (C.I.)	Frequency (f)	Mid. point of C.I. (X)	fX	x = X - M	x	f x
1-10	3	5.5	16.5	-42	42	126
11-20	5	15.5	77.5	-32	32	160
21-30	6	25.5	153.0	-22	22	132
31-40	4	35.5	142.0	-12	12	48
41-50	6	45.5	273.0	-2	2	12
51-60	12	55.5	666.0	8	8	96
61-70	6	65.5	393.0	18	18	108
71-80	4	75.5	302.0	28	28	112
81-90	3	85.5	256.5	38	38	114
91-100	1	95.5	95.5	48	48	48
	N=50		$\sum fX = 2375$			$\sum f x = 956$

- iv) Each frequency is to be multiplied by the absolute deviation of the mid-point of its C.I. and the product ($f|x|$) is recorded in the seventh column.
- v) Sum of $f|x|$ is to be computed.
- vi) M.D. is computed by using the formula $M.D. = \frac{\sum f|x|}{N}$

In our example $M.D. = \frac{956}{50} = 19.12$

5:10:4:04 Merits and Demerits of Mean Deviation

Merits:

M.D. is based on mean and hence all items are taken into account.

Demerits:

- It is less stable than S.D.
- The signs of the deviations are neglected and hence it is unsuitable for further algebraic treatment.
- It is not based on sound mathematical formula.

5:10:5 Standard Deviation (S.D.)

Standard Deviation (S.D) is the square-root of the mean of the squares of the deviations of all the items from the Arithmetic Mean [Root of Mean of Squared Deviations – R.M.S.Deviations]. It is represented by the Greek letter 'σ'.

In the computation of deviation of each score in the given data, from the mean (i.e. $\bar{x}$) some may be

positive and some may be negative. When we add all the deviations to find the average, it always turns out to be zero [i.e. we are unable to calculate M.D.]. To avoid this, the absolute value of each deviation is to be found out by dropping the signs. But this is not a proper mathematical procedure. Therefore a way out from this problem is found out in the computation of S.D. i.e. each deviation is squared and the sum of the squared deviations ($\sum x^2$) is divided by the total number of items N . This is called 'Variance' ($\frac{\sum x^2}{N}$). The square root of this variance gives us the 'Standard Deviation'.

For raw data, the formula is $\sigma = \sqrt{\frac{\sum x^2}{N}}$

Where $x = X - \bar{X}$

It can also be calculated by using the formula

$$\sigma = \sqrt{\frac{\sum d^2}{N} - \left(\frac{\sum d}{N}\right)^2}$$

Where $d = X - A$ and A is the assumed mean.

Square of Standard Deviation is known as **variance**.

Formula for computing S.D for a frequency distribution is

$$\sigma = i \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2}$$

Where $d = \frac{X - A}{i}$ and i = class size.

5:10:5:01 Computation of Standard Deviation for Ungrouped Data

Illustration :

Find S.D. for the following data :

16, 14, 24, 19, 17, 18

Steps in finding S.D.

- i) Add the values X and divide it by N to get the Mean (M).

- ii) Find the deviation of each item from the mean (i.e. x). Check whether $\Sigma x = 0$

- iii) Square each value of x [Find x^2]

X	$x = X - M$	x^2
16	-2	4
14	-4	16
24	6	36
19	1	1
17	-1	1
18	0	0
$\Sigma X = 108$ $\Sigma x = 7-7$ $\Sigma x^2 = 58$		
$M = \Sigma X / N$ $= 0$ $\sigma = \sqrt{58/6} = 3.109$		
$= 108/6$		
$= 18$		

- iv) Add all values of x^2 [i.e. Σx^2]

- v) Find the average of Σx^2 (i.e. $\Sigma x^2 / N$) which is called 'variance'

- vi) Find the square root of the variance (i.e. $\sqrt{\Sigma x^2 / N}$)

Note: If the mean involves fraction, then this method becomes difficult. In that case the following method (Using Assumed Mean) is employed for easy computation.

Example :

X	$d = X - A$	d^2
16	-3	9
14	-5	25
24	5	25
A (19)	0	0
17	-2	4
18	-1	1
$\Sigma d = -11 + 5$ $\Sigma d^2 = 64$		
$= -6$		

$$\sigma = \sqrt{\frac{\Sigma d^2}{N} - \left[\frac{\Sigma d}{N}\right]^2}$$

$$= \sqrt{\frac{64}{6} - \left[\frac{-6}{6}\right]^2}$$

$$= \sqrt{10.67 - 1} = \sqrt{9.67} = 3.109$$

[Note: Students are advised to use this short cut method always. Further any item in the group of data can be used as 'Assumed Mean']

5:10:5:02 Computation of S.D. for Grouped Data (Discrete series)

Illustration :

Find the S.D. for the following data :

Marks obtained :	20	25	30	35	40
No. of students :	4	10	14	6	6

Solution

Marks (X)	No. of Students (f)	$d = X - A$	fd	$fd^2 = fd \times d$
20	4	-10	-40	400
25	10	-5	-50	250
A (30)	14	0	0	0
35	6	5	30	150
40	6	10	60	600
$N = \Sigma f = 40$		$\Sigma d = 0$	$\Sigma fd = -90 + 90 = 0$	$\Sigma fd^2 = 1400$

$$\sigma = \sqrt{\frac{\Sigma fd^2}{N} - \left[\frac{\Sigma fd}{N}\right]^2}$$

$$= \sqrt{\frac{1400}{40} - \left[\frac{0}{40}\right]^2}$$

$$= \sqrt{35 - 0} = \sqrt{35} = 5.916$$

$$\Omega \ 5.92$$

Steps in the Procedure :

- Take any item, preferably that one at the middle as the assumed Mean A. [In our example 'A' has been assigned the value 30].
- From each value found in the first column, subtract 'A' and write the deviation 'd' in column No. 3.
- Multiply each frequency (found in column 2) by its respective deviation (found in column 3) and record the products (i.e. fd) in column 4.
- Multiply each value in column 3 and that in column 4 and record the product fd^2 in column 5.
- Compute the S.D. using the formula

$$\sigma = \sqrt{\frac{\sum fd^2}{N} - \left[\frac{\sum fd}{N} \right]^2}$$

5:10:5:03 Computation of S.D. for Grouped Data (Continuous Series)

Taking the frequency distribution given in 5:09:1:02 as an example, the short-cut method of computing S.D. is described as follows :

Illustration :

Class Interval (C.I)	Frequency (f)	Mid-Point of C.I. (X)	Step Deviation $d = \frac{X - A}{i}$	'fd'	(fd ²)
1-10	3	5.5	-5	-15	75
11-20	5	15.5	-4	-20	80
21-30	6	25.5	-3	-18	54
31-40	4	35.5	-2	-8	16
41-50	6	45.5	-1	-6	6
51-60	12	55.5	0	0	0
61-70	6	65.5	1	6	6
71-80	4	75.5	2	8	16
81-90	3	85.5	3	9	27
91-100	1	95.5	4	4	16
$i=10$	$N = \sum f = 50$			$\sum fd = -67 + 27 = -40$	$\sum fd^2 = 296$

Steps in the procedure :

- As in the computation of mean by the short-cut method, write the step deviation $d = \frac{X - A}{i}$
- Find fd and $\sum fd$ (as is done in the case of computation of mean by short-cut method)
- For finding ' fd^2 ', multiply each ' fd ' in column 5 by the corresponding ' d ' value found in column 4.
- Find the sum of fd^2 values . i.e. find $\sum fd^2$
- Use the formula $\sigma = i \sqrt{\frac{\sum fd^2}{N} - \left[\frac{\sum fd}{N} \right]^2}$ by substituting the respective values.

$$\begin{aligned}
 \text{In our example } \sigma &= 10 \times \sqrt{\frac{296}{50} - \left[\frac{-40}{50}\right]^2} \\
 &= 10 \times \sqrt{5.92 - .64} \\
 &= 10 \times \sqrt{5.28} \\
 &= 10 \times 2.298 \\
 &= 22.98
 \end{aligned}$$

5:10:5:04 Situations Where S.D. is Used

- When a highly reliable measure of dispersion is needed.
- Where mean is used as a measure of 'average', then standard deviation alone should be used as a measure of dispersion.
- To interpret the distributions based on Normal Probability Curve, standard deviation is highly useful.
- To compute and compare the skewness of distributions, S.D. is used.
- To compare the dispersion or scatteredness of distributions, 'variance' is used which involves S.D.
- S.D. is used extensively in Statistical Quality Control methods.

5:10:6 Quartile Deviation (Q.D.)

To divide a distribution in which items are arranged in ascending order, into 4 equal parts, three points are

required and these three points are called 'Quartile points'.

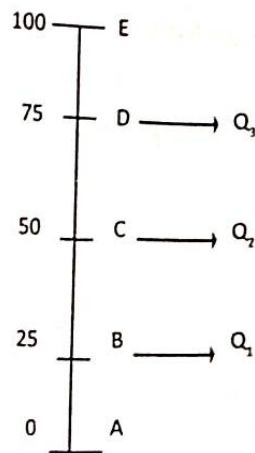
In the given figure, Point 'B' divides the scale in the ratio 1:3; i.e. below the point 'B', $\frac{1}{4}$ of the total items (i.e. 25% of the items) lie. This point is called the '**First Quartile**' or '**Lower Quartile**' and denoted by the letter Q_1 .

Below the point C, the second quartile point, 50% of the total items lie. The second quartile point is otherwise known as '**Median**' and is denoted by the letter ' Q_2 '.

Similarly, 'D' is the third quartile (or upper quartile) point below which $\frac{3}{4}$ of the total items (i.e. 75% of the items) lie and this point is denoted by Q_3 .

The difference between the upper and lower quartiles namely Q_3 & Q_1 is called '**Inter Quartile Range**' and half of this is known as '**Quartile Deviation**'. Hence we can say Q.D is the '**semi-inter-quartile range**'.

$$Q.D. = \frac{Q_3 - Q_1}{2}$$



5:10:6:01 Computation of Quartile Deviation for Ungrouped Data

Illustration:

Calculate Q.D. for the following 30, 90, 20, 10, 80, 70

Step 1: Rearranging the data in the ascending order

10, 20, 30, 70, 80, 90

Step 2: $Q_1 = \text{Value of } \frac{(N+1)}{4} \text{ th item}$

Here $N = 6$

$$\therefore Q_1 = \frac{7}{4} = 1.75^{\text{th}} \text{ term}$$

= Value of the item $1 + 0.75 \times (\text{Value of the item 2} - \text{Value of the item 1})$

$$= 10 + 0.75 (20 - 10)$$

$$= 10 + 7.5$$

$$= 17.5$$

Step 3: $Q_3 = \text{Value of the } 3 \frac{(N+1)}{4} \text{ th item}$

Here $N = 6$

$$\therefore Q_3 = 3 \left(\frac{6+1}{4} \right) = \frac{21}{4} = 5.25^{\text{th}} \text{ item}$$

= Value of the 5th item $+ 0.25 (\text{Value of the 6th item} - \text{Value of the 5th item})$

$$= 80 + 0.25 (90 - 80) = 80 + .25 (10) = 82.5$$

$$\text{Step 4: } Q = \frac{Q_3 - Q_1}{2} = \frac{82.5 - 17.5}{2} = \frac{65}{2} = 32.5$$

5:10:6:02 Computation of Q.D. for Grouped Data (Discrete Series)

Illustration:

Marks obtained 10 20 30 60 70 80 90

No. of Students 50 40 30 40 10 20 30

Solution

Marks obtained (X)	No. of Students (f)	Cumulative Frequencies (C.f)
10	50	50
20	40	90
30	30	120
60	40	160
70	10	170
80	20	190
90	30	220
Total	$N = \sum f = 220$	

Steps in the Procedure

- Form the table such that value of items are placed in the ascending order.
- Compute & Write the cumulative frequencies in Column 3 of the table.

iii) Computing the first Quartile

The first quartile lies in the interval in which $\frac{N}{4}$ falls

$$\text{Here } \frac{N}{4} = \frac{220}{4} = 55$$

55th item lies between the score values 10 and 20. The lower limit l_1 of this is, 10, $C_1=50$, $f_1=40$, $i=10$.

Substituting these values of in the formula for first quartile

$$\begin{aligned} Q_1 &= l_1 + \left[\frac{\frac{N}{4} - C_1}{f_1} \right] \times i \\ &= 10 + \frac{55 - 50}{40} \times 10 \\ &= 10 + \left[\frac{5 \times 10}{40} \right] = 10 + \frac{50}{40} \\ &= 10 + 1.25 = 11.25 \end{aligned}$$

iv) Computing the Third Quartile

Third quartile lies in the interval, where $\frac{3N}{4}$ falls.

Here $\frac{3N}{4} = \frac{3 \times 220}{4} = 165$ th item, which lies in the interval between scores 60 and 70. Here $l_3=60$; $C_3=160$; $f_3=10$; $i=10$.

Substituting these values in the formula for third quartile

$$\begin{aligned} Q_3 &= l_3 + \left[\frac{\frac{3N}{4} - C_3}{f_3} \right] \times i \\ &= 60 + \left[\frac{165 - 160}{10} \right] \times 10 \end{aligned}$$

$$= 60 + \frac{5 \times 10}{10} = 60 + 5 = 65$$

$$\begin{aligned} \text{v) } Q.D. &= Q = \frac{Q_3 - Q_1}{2} = \frac{65 - 11.25}{2} \\ &= \frac{53.75}{2} = 26.875 \approx 26.88 \end{aligned}$$

5:10:6:03 Computation of Q.D. for Grouped Data (Continuous Series)

Illustration

Taking the frequency distribution given in 10:4:01:2 as an example, we can workout the solution as follows :

Steps in the Procedure :

Step I - Finding Q_1

- a) First quartile class is the class in which falls. Here $\frac{N}{4} = \frac{50}{4} = 12.5$ th item which falls in the C.I. 21-30.

$$\text{b) } Q_1 = l_1 + \left[\frac{\frac{N}{4} - C_1}{f_1} \right] \times i \quad \text{where}$$

l_1 = Exact lower limit of the first quartile class

Here $l_1 = 20.5$

Class Interval (C.I)	Frequencies (f)	Cummulative Frequencies (C.f)
1-10	3	3
11-20	5	8
21-30	6	14 $\leftarrow N/4$
31-40	4	18
41-50	6	24
51-60	12	36
61-70	6	42 $\leftarrow 3N/4$
71-80	4	46
81-90	3	49
91-100	1	50
$i=10$	$N=\Sigma f=50$	

C_1 = Cumulative frequency, just before the first quartile class = 8

f_1 = frequency in the first quartile class = 6

$$Q_1 = 20.5 + \left[\frac{12.5 - 8}{6} \times 10 \right]$$

$$= 20.5 + \left[\frac{4.5}{6} \right] \times 10$$

$$= 20.5 + 45/6 = 20.5 + 7.5 = 28$$

Step II - Finding Q_3

a) Third quartile class is the class in which $\frac{3N}{4}$ falls.
Here $\frac{3N}{4} = \frac{3 \times 50}{4} = 37.5$ th items which falls in the C.I. 61-70.

$$b) Q_3 = l_3 + \left(\frac{\frac{3N}{4} - C_3}{f_3} \right) \times i$$

Here l_3 = The exact lower limit of the third quartile class = 60.5

C_3 = Cumulative frequency just before the third quartile class = 36

f_3 = frequency of the third quartile class = 6

Therefore in our example,

$$Q_3 = 60.5 + \left(\frac{37.5 - 36}{6} \times 10 \right) \Rightarrow 60.5 + \left(\frac{1.5 \times 10}{6} \right) \Rightarrow 60.5 + 2.5 = 63$$

Step III - Finding the value of Q.D.

$$Q.D. = Q = \frac{Q_3 - Q_1}{2} = \frac{63 - 28}{2} = \frac{35}{2} = 17.5$$

5:11 Correlation

5:11:1 Concept of Correlation

Measures of central tendency and dispersion tell us about the nature of a given variable, whereas correlation states the relationship between two or more variables. For example, according to the height of students, their weight also changes. Therefore, between the two variables 'height' and 'weight', there exists a relationship. Similarly there exists a relationship between 'price' of goods and the 'demand' for them. From the point of view of statistics, **two variables are said to be related if the changes in one variable are accompanied by changes in the other variable.**

The degree or amount of relationship existing between two variables is measured by what is known as "**Correlation Coefficient**" whose value ranges between -1 and +1.

5:11:2 Types of Correlation

Depending on the nature of relationship existing between variables, correlation is of following types.

1. Positive and negative correlation.
2. Simple, Partial and multiple Correlations.
3. Linear and Curvilinear Correlation.

5:11:2:01 Positive and Negative Correlations

Positive Correlation

Variables are said to be positively related if change in one variable brings forth change in the other variable in the same direction (i.e. if the value of one variable increases the value of other variable also increases; if the value of one variable decreases, the value of other variable also decreases).

Examples

- i) Marks obtained by students in Mathematics and Physics
- ii) Intelligence and Achievement scores
- iii) Height and Weight

Negative Correlation

Variables are said to have negative or inverse correlation, when the value of one variable changes, change in the value of the other variable also takes place but in the opposite direction.

Examples

1. Speed and the time required to cover a given distance.
2. Pressure and volume of gas in a closed container.
3. Prices of goods and the demand for them.

Further, positive correlation is of two types: (i) Perfect positive correlation and (ii) Imperfect positive correlation. Coefficient of correlation for

positive perfect correlation between two variables is +1 [E.g. Speed and distance covered]. The value of imperfect correlation is a fraction between 0 and 1 [E.g. Height and weight, intelligence and achievement etc.]. Similarly negative correlation is of two types : (i) Perfect negative correlation and (ii) Imperfect negative correlation. The value of 'Perfect negative correlation' is -1 [E.g. Speed and time taken to cover a given distance, pressure and volume of a given mass of a gas etc.]. The value of imperfect negative correlation is a fraction between 0 to -1 [E.g. No of students in the class and the amount of individual attention paid by the teacher, prices of goods and the demand for them etc.]

Zero Correlation or Absence of Correlation

Two variables are said to be unrelated or independent of each other, if the changes in one variable do not affect the other. [E.g.. Intelligence and weight, height etc. and academic achievement, proficiency in music and mathematics etc.]

5:11:3 Measuring the Correlation between Two Variables

If the values of any two variables say X and Y are given, then based on these values, to find the nature and extent of relation between them could be found using either of the following two

- i) Karl Pearson's Product Moment Correlation Coefficient

ii) Spearman's Rank Order Correlation Coefficient

I Karl Pearson's Product Moment Correlation Coefficient $r = \frac{\sum xy}{N\sigma_x\sigma_y}$

Here X, Y are the variables; $x = X - \bar{X}$, $y = Y - \bar{Y}$, $\bar{X}$ and $\bar{Y}$ are the means of X and Y values respectively. σ_x , σ_y are the Standard Deviations of the X and Y values, of X and Y variables; N is the total pairs of scores. In the above formulae, if we substitute the formula for the S.Ds, then $r = \frac{\sum xy}{\sqrt{\sum x^2} \sqrt{\sum y^2}}$

But in practice we mostly make use of assumed means. In that case

$$\text{correlation coefficient } r = \frac{\sum xy - \sum x \sum y}{\sqrt{[N\sum x^2 - (\sum x)^2][N\sum y^2 - (\sum y)^2]}}$$

Where N = Total pairs of scores given.

x = Deviation of X values of the Variable X from the assumed mean (A).

$$\text{i.e. } x = X - A$$

Similarly $y = Y - B$ where 'B' is the assumed mean for the values of Y variable.

II Spearman's Rank Order Correlation Coefficient $\rho = 1 - \frac{6\sum D^2}{N(N^2-1)}$ (rho)

Where D = Difference in ranks of a pair values of X & Y respectively i.e. $D = R_x - R_y$

N = Total number of pairs of values given.

5:11:3:01 Computing Karl Pearson's Correlation Coefficient for Ungrouped Data

Illustration : Compute the correlation coefficient 'r' for the following data.

X: 32, 42, 26, 63, 38, 45, 45, 56, 62, 26, 52, 64, 39, 91, 18, 24

Y: 62, 36, 19, 72, 44, 54, 50, 62, 76, 36, 54, 24, 45, 64, 36, 54

X: 32, 42, 26, 63, 38, 45, 45, 56, 62, 26, 52, 64, 39, 91, 18, 24

Y: 62, 36, 19, 72, 44, 54, 50, 62, 76, 36, 54, 24, 45, 64, 36, 54

Marks in History (X)	Marks in Geogra - phy(Y)	$x' = X - A$	$y' = Y - B$	$x'y'$	$(x')^2$	$(y')^2$
34	62	-22	+12	-264	484	144
42	36	-14	-14	196	196	196
26	19	-30	-31	930	900	961
63	72	7	22	154	49	484
38	44	-18	-6	108	324	36
45	54	-11	4	-44	121	16
45	B 50	-11	0	0	121	0
A 56	62	0	12	0	0	144
62	76	6	26	156	36	676
26	36	-30	-14	420	900	196
52	54	-4	4	-16	16	16
64	24	8	-26	-208	64	676
39	45	-17	-5	85	289	25
91	64	35	14	490	1225	196
18	36	-38	-14	532	1444	196
24	54	-32	4	-128	1024	16

Assumed mean of X=56 $\sum x' =$ $\sum y' =$ $\sum x'y' =$ $\sum (x')^2 =$ $\sum (y')^2 =$

Assumed mean of Y=50 -171; -12; 2411; 7193; 3978

Here N = 16

$$\begin{aligned}
 &= \frac{N \sum x'y' - \sum x' \sum y'}{\sqrt{[N \sum x'^2 - (\sum x')^2][N \sum y'^2 - (\sum y')^2]}} \\
 &= \frac{16(2411) - (-171)(-12)}{\sqrt{[16(7193) - (-171)^2][16(3978) - (-12)^2]}} \\
 &= \frac{38576 - 2052}{\sqrt{(115088 - 29241)(63648 - 144)}} \\
 &= \frac{36524}{\sqrt{85847 \times 63504}} = \frac{36524}{\sqrt{85847} \sqrt{63504}} \\
 &= \frac{36524}{292.996 \times 252} = .4946 \approx .50
 \end{aligned}$$

Steps in the Procedure of the Solution

- Write the values of X and Y in columns 1 and 2 respectively.
- Take any value of X as the assumed mean and subtract it from every value of X and write the deviations thus obtained (x') in Column 3. [In the illustration given, assumed mean value is taken as 56.]
 - Compute the sum of x' values i.e. $\sum x'$ [In our example = $\sum x' = -171$]
- Take any value of Y as the assumed mean and subtract it from every value of Y and write the deviations thus obtained (y') in column 4 [In the illustration given, assumed mean of Y values is taken as 50 (denoted as B)]
 - Compute the sum of y' values i.e. $\sum y'$ [In our example $\sum y' = -12$]

- In our example $\sum x'y' = 2411$ Multiply each x' and y' ; write the products in column 5 and find the sum of $x'y'$.
- Write the square of each x' [i.e. $(x')^2$] in column 6. Find the sum of $(x')^2$ [In our example $\sum (x')^2 = 7193$]
- Write the square of each y' [i.e. $(y')^2$] in column 7. Find the sum of $(y')^2$. [In our example $\sum (y')^2 = 3978$]
- $$r = \frac{N \sum x'y' - \sum x' \sum y'}{\sqrt{[N \sum (x')^2 - (\sum x')^2][N \sum (y')^2 - (\sum y')^2]}}$$

Substituting the respective values in the above formulae i.e.

$$N = 16; \sum x'y' = 2411; \sum x' = -171; \sum y' = -12$$

$$\sum x'^2 = 7193; \text{ and } \sum y'^2 = 3978$$

$$\begin{aligned}
 \therefore r &= \frac{16(2411) - (-171)(-12)}{\sqrt{16(7193) - (-171)^2} \sqrt{16(3978) - (-12)^2}} \\
 &= \frac{38576 - 2052}{\sqrt{115088 - 29241} \sqrt{63648 - 144}} \\
 &= \frac{36524}{\sqrt{85847} \sqrt{63504}} = \frac{36524}{292.996 \times 252} = .4946 \approx .50
 \end{aligned}$$

5:11:3:02 Computing Rank Order Correlation Coefficient

This method can be illustrated with the following example.

X : 91, 13, 65, 46, 54, 46, 46, 65, 39, 48

Y : 82, 31, 70, 42, 62, 34, 71, 62, 42, 46

X	Y	Ranks in X (R_x)	Ranks in Y (R_y)	Rank Difference $D=R_x-R_y$	D^2
91	82	1.0	1.0	0	0
13	31	10.0	10	0	0
65	70	2.5	3	-.5	.25
46	42	7.0	7.5	-.5	.25
54	62	4.0	4.5	-.5	.25
46	34	7.0	9	-2	4.0
46	71	7.0	2	5	25.0
65	62	2.5	4.5	-2	4.0
39	42	9.0	7.5	1.5	2.25
48	46	5.0	6	-1	1.00

N=10

$$\begin{aligned}\sum D &= 6.5 - 6.5 \\ &= 0\end{aligned}$$

$$\sum D^2 = 37$$

Steps in the Procedure for the Solution

- Ranks should be assigned for each of the values of the variables. Among the values of a variable, the highest value should be given the rank 1 and the succeeding lower values should be given ranks 2, 3 and so on.
- If two or more items get the same score value, then thinking each one is given a separate rank, the average of the ranks involved should be found out and assigned to all the items having the same score value. For example, in our illustration the highest X value 91 gets the rank 1. The next two ranks (namely 2 and 3) are to be shared by the values 65. We assign the 'tied-up rank' $2.5 \left[\frac{2+3}{2} = 2.5 \right]$ to both the scores 65. The next X value 54 gets the 4th rank. In this way, assigning a rank for each

item should proceed. In our illustration after assigning the rank 5 for the X value 48, there are three items having the same value 46. For them, thinking ranks 6, 7 and 8 are given, the average of these ranks is calculated as $\frac{6+7+8}{3} = 7$. Hence, these three items having the same value 46 are assigned the 'tied-up rank' 7.

- The difference between the ranks of a pair of items ($D=R_x-R_y$) is calculated and written in column 5. Sum of D [i.e. $\sum D$] is always equal to zero (check).
- The square of the difference in ranks of each pair of X & Y values [i.e. D^2] is calculated and written in column 6. The sum of these squared differences in ranks ($\sum D^2$) is to be computed.
- Using the formula, Spearman's rank order correlation is computed.

In our illustration $\sum D^2 = 37$; $N=10$

$$\rho = 1 - \frac{6(37)}{10 \times 99}$$

$$= 1 - \frac{6(37)}{10(100-1)}$$

$$= 1 - \frac{222}{990} = 1 - .224$$

$$= .776 \text{ (High Positive Correlation)}$$

5:11:4 Interpretation of Correlation Coefficient

- The value of correlation is always between +1 and -1.

2. If $r = 1$ then there is perfect positive correlation between the variables. If $r = -1$, then there is perfect negative correlation between variables. Generally the obtained value of 'r' will be a fraction.

± 0.81 to ± 1.0 represents Very high level of relationship.

± 0.61 to ± 0.80 represents High level of relationship.

± 0.41 to ± 0.60 represents Moderate or significant level of relationship.

± 0.21 to ± 0.40 represents low level of relationship.

0 to ± 0.20 represents very low or insignificant level of relationship.

3. $r = -0.6$ indicates negative or inverse relationship between variables and the amount of relationship is 0.6 (Moderate).

5:11:5 Uses of Product Moment Correlation Coefficient

- It helps to find out the kind as well as the degree of relationship that may exist between two sets of scores.
- It is used to establish the reliability and validity of tests.
- It is used in regression equations and factor analysis.

5:12 Graphs and Diagrams

The most important function of statistics is to present the complex data in an easy to understand manner. For this purpose, data are classified and tabulated; to present easy comparisons, diagrams and graphs are also used. Some of the important graphic

representations used in statistics are: (i) Bar Diagram, (ii) Pie Diagram, (iii) Histogram, (iv) Frequency Polygon, (v) Cumulative Frequency Curve and Ogive (Percentage Cumulative Curve). These figures enable us to understand at a glance, how the frequencies are distributed in the various class intervals, giving an idea of the essential nature of the distribution.

5:12:1 Diagrams

5:12:1:01 Bar Diagram

In statistics, bar diagrams are extensively used as they are easy to draw and help for easy comparisons. Bar Diagrams are used (i) to compare the different persons or groups on a single variable and (ii) to compare a person or group on many variables.

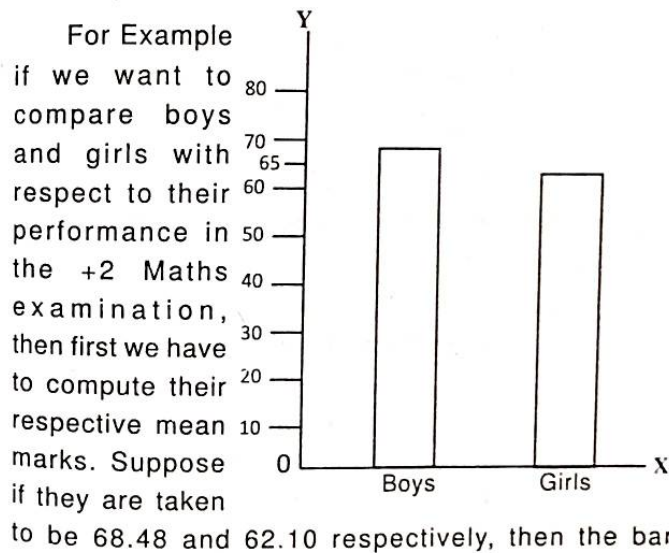
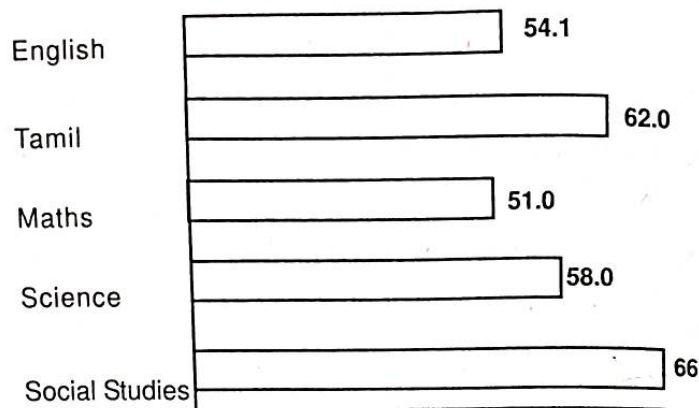


diagram will be as shown on the right side. Bar diagrams can also be drawn, using horizontal bars. For example the bar diagram presented below could be used to compare the mean performance of students in different school subjects.



Following are the general rules to be followed while drawing Bar Diagrams:

1. Bars are to be drawn from a common base line.
2. All bars are to be of equal width
3. The gap between any two bars is to be equal.
4. For Vertical Bars, height of each bar should be measured on the Y-Axis and for the horizontal Bar, the length of each bar should be made on the X-Axis
5. Foot notes are to be written legibly. For Vertical Bars foot notes should be written below the Bar line and for Horizontal bars, footnotes should be on the left of the Bar line.

6. Similar items are to be represented in the bar diagram with the same colour, dots or lines.

5:12:1:02 Pie Diagram

In Pie Diagram, total frequency of a distribution is taken to represent the area of the circle drawn and the different sectors of the circle are according to the frequencies of the different categories of the distribution. Pie-Diagram helps to understand the comparison of statistical data in the visual form.

Total angle at the centre is 360° . Hence according to the size of the frequencies of the different categories in the distribution, angle at the centre is apportioned and accordingly the sectors are drawn. To differentiate the sectors, different colours or designs could be used. A note indicating what each colour or design represents should also be provided by the side of the Pie Diagram.

Illustration :

Draw a Pie-Diagram for the following monthly budget of a family.

Categories of expenses	House Rent	Food	Health Care	Educa-tion	Savings	Total
Expense Amount (In Rupees)	5,000	8,000	1,000	3,000	2,000	20,000

Solution

Steps in the Procedure :

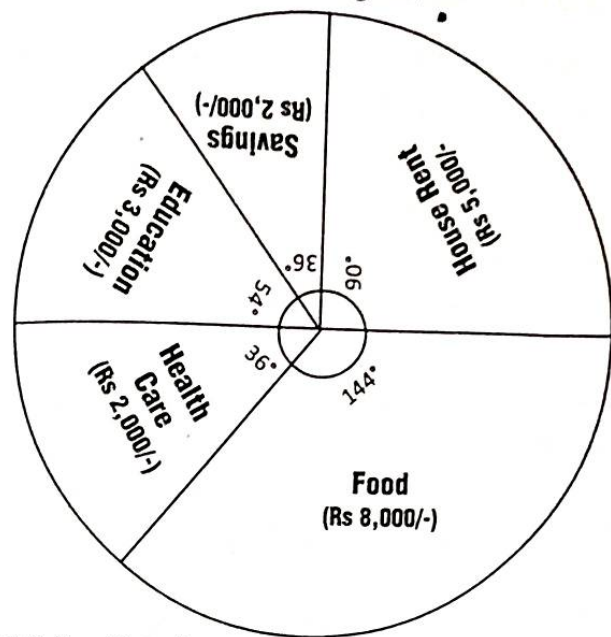
- i) Draw a circle with a suitable radius.

- ii) Each category of expenses is to be divided by total amount of the budget and the result is multiplied by 360 to find the sector angle for each category of expense (i.e. $\frac{f}{N} \times 360$ where f = category of expense; N = Total budget amount).
- iii) As per the categories of expenses sectors are to be formed in the circle.
- iv) To differentiate each sector, different colours or designs should be used.
- v) Note indicating what each colour or design represents, should be marked by the side of the Pie-Diagram

Calculating the angle of sector in the circle for Different Categories of Expenses

Categories of expenses	Expenses Amount (in Rs.)	Sector angle
House Rent	5,000	$\frac{5000}{20,000} \times 360^\circ = 90^\circ$
Food	8,000	$\frac{8000}{20,000} \times 360^\circ = 144^\circ$
Health Care	2,000	$\frac{2000}{20,000} \times 360^\circ = 36^\circ$
Education	3,000	$\frac{3000}{20,000} \times 360^\circ = 54^\circ$
Savings	2,000	$\frac{2000}{20,000} \times 360^\circ = 36^\circ$
Total	20,000	360°

Pie Diagram



5:12:2 Graphs

There are diagrams drawn on graph sheets. A frequency distribution can be presented graphically in any one of the following ways.

1. Histogram
2. Frequency Polygon
3. Cumulative Frequency Curve
4. Ogive Curve (or Cumulative Frequency Percentage Curve)

Let us see each one in detail in the following sections.

5:12:2:01 Histogram

Histogram is the most popular and widely used graph for presenting the given data in the form of a frequency distribution. In this kind of graphical presentation, the exact limits of class intervals are represented on the X - axis and frequencies on the Y - axis respectively on suitable scales. The base of each rectangle represents the width of the class interval and the height represents the frequency in that class interval. The area of each rectangle is proportionate to the frequency of the corresponding class. The frequency is supposed to spread uniformly over the entire area of the rectangle. The adjacent rectangles thus formed represents Histogram.

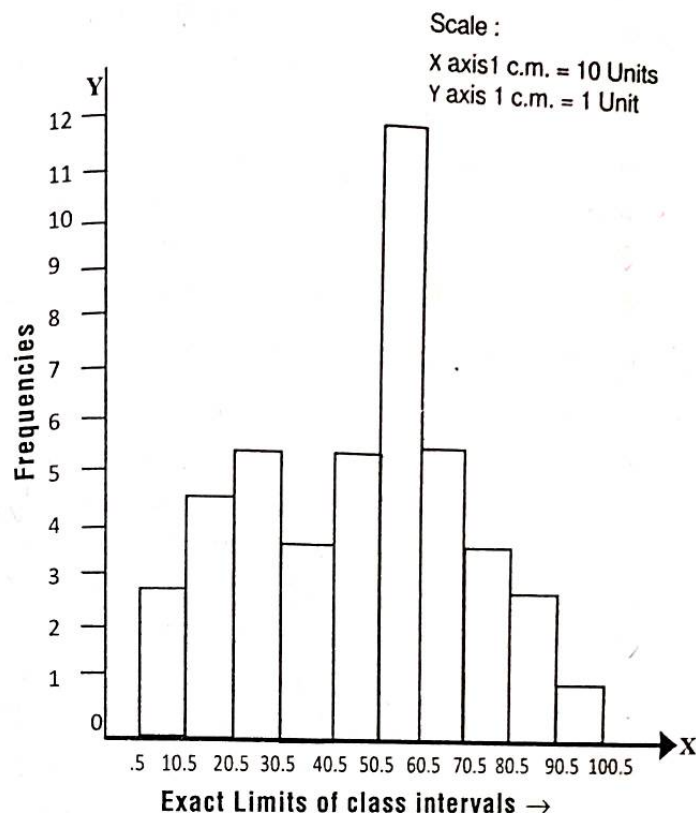
At a glance, we can understand the nature of a given frequency distribution with the help of a histogram; but it is not useful to compare two or more frequency distributions.

Illustration

Draw a 'Histogram' for the frequency distribution given.

C.Is	1-10	11-20	21-30	31-40	41-50	51-60	61-70	71-80	81-90	91-100
Frequencies	3	5	6	4	6	12	6	4	3	1

Solution



5:12:2:02 Frequency Polygon

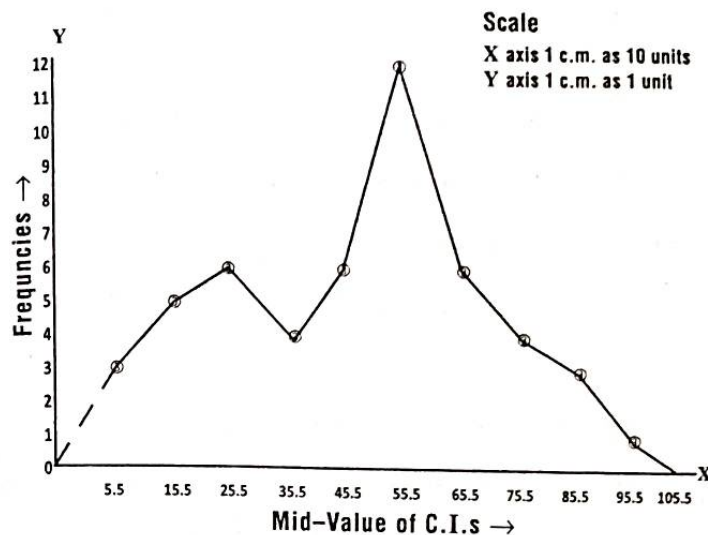
In frequency polygon, the frequency of each class interval is supposed to be concentrated at the mid-point of the respective class interval. Hence to draw the frequency polygon,

- Mark the exact score limits in the X-Axis on a suitable scale.

- ii) Take the frequencies on the Y-Axis on a suitable scale
- iii) Mark the frequency of each class interval as a point, directly above the mid-point of the concerned class. For example in the illustration we have taken (the frequency distribution as used in histogram), we can mark the points (5.5, 3), (15.5, 5), (25.5, 6), (35.5, 4), (45.5, 6), (55.5, 12), (65.5, 6), (75.5, 4), (85.5, 3), and (95.5, 1) on the graph paper.
- iv) Join the points with straight lines.
- v) To complete the frequency polygon, its ends are to be joined at 0 and 105.5 respectively in the following manner. The mid-point of the class interval before the C.I (1-10) is 0. Similarly the mid-point of the C.I. next to the C.I (91-100) is also 0. Therefore the left and right ends of the figure drawn should be joined to the X axis at (0, 0) and (105.5, 0) respectively. The closed figure thus obtained is the 'Frequency Polygon'. The area enclosed by the frequency polygon represents the total frequency of the distribution. To represent two or more frequency distributions as frequency polygons drawn on the same graph sheet, instead of frequencies, percentage of frequencies are to be taken in the Y - axis. To compare two or more frequency distributions, frequency polygon is the best suited one.

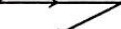
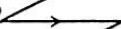
Illustration

Frequency Polygon for the frequency distribution given in 5:09:1:02 is as follows :



5:12:2:03 Cumulative Frequency Curve

Cumulative frequency (c.f.) for a particular class interval (C.I.) is the sum of the frequency of the C.I. and the frequencies of all class intervals below it, having lesser values as their lower and upper limits i.e. cumulative frequency is obtained by starting with the C.I. having the lowest marks and then adding the frequencies of successive C.Is. In our example, for the frequency distribution given below, the method of obtaining the cumulative frequencies is indicated with arrow marks.

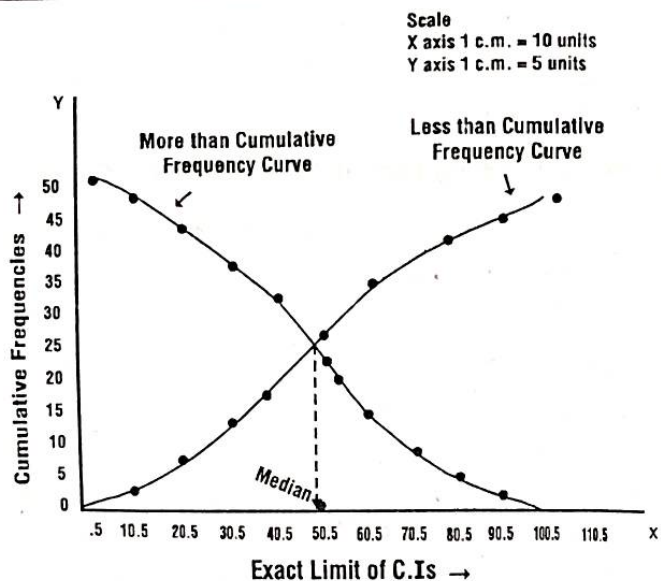
CI	f	Cumulative frequency (cf)	
		'Less than' Cumulative frequency	'More than' Cumulative frequency
1-10	3 	3	50 (47+3)
11-20	5 	8	47 (42+5)
21-30	6 	14	42 (36+6)
31-40	4 	18	36 (32+4)
41-50	6 	24	32 (26+6)
51-60	12 	36	26 (14+12)
61-70	6 	42	14 (8+6)
71-80	4 	46	8 (4+4)
81-90	3 	49	4 (1+3)
91-100	1 	50	1 (0+1)

[For calculating the 'More than' cumulative frequencies, we start from the C.I. having the highest values as its lower and upper limits and then go on adding the frequencies of successive class intervals. Cumulative frequencies are useful to compute how many items are there above the lower limit or below the upper limit of a class interval. For example in the distribution given above, the exact upper limit of the class interval 31-40 is 40.5, below which there are 18 pupils (4+6+5+3) as indicated by the 'Less than' cumulative frequency. Similarly for the class interval

51-60, the exact lower limit is 50.5 above which 26 pupils (12+6+4+3+1) have their scores, as indicated by the 'More than' cumulative frequency.

Method of Drawing Cumulative Frequency Curve

- Mark the exact limits of C.Is on the X-axis and the cumulative frequencies on the Y-axis with suitable scales.
- Above the exact upper limit of each class interval, mark the point representing the 'less than' cumulative frequency corresponding to the class. The points thus marked [Such as (10.5, 3), (20.5, 8), (30.5, 14), (80.5, 46), (90.5, 49), (100.5, 50)] are to be joined by free hand. The left end of this curve should be joined to the exact upper limit of the class previous to the one given in the distribution with lowest values as its lower and upper limits. In our example it turns out to be (0, 0). The Curve thus obtained is "**Less than Cumulative Frequency Curve**".
- For drawing the "**More then Cumulative Frequency Curve**", the procedure is the same as described in step (ii), but above the exact lower limit of each class interval, the point representing the 'more than' cumulative frequency corresponding the class is to be marked.
- These two cumulative frequency curves intersect each other at a point. The X - coordinate of this point of intersection is the median value.



5:12:2:04 Ogive Curve (or Cumulative Percentage Frequency Curve)

If percentage cumulative frequencies are taken on the Y-axis and the exact limits of class intervals on the X-axis, the frequency curve thus drawn is called 'ogive'.

$$\text{Percentage cumulative frequency} = \left[\frac{100}{N} \right] \times \text{Cumulative Frequency}$$

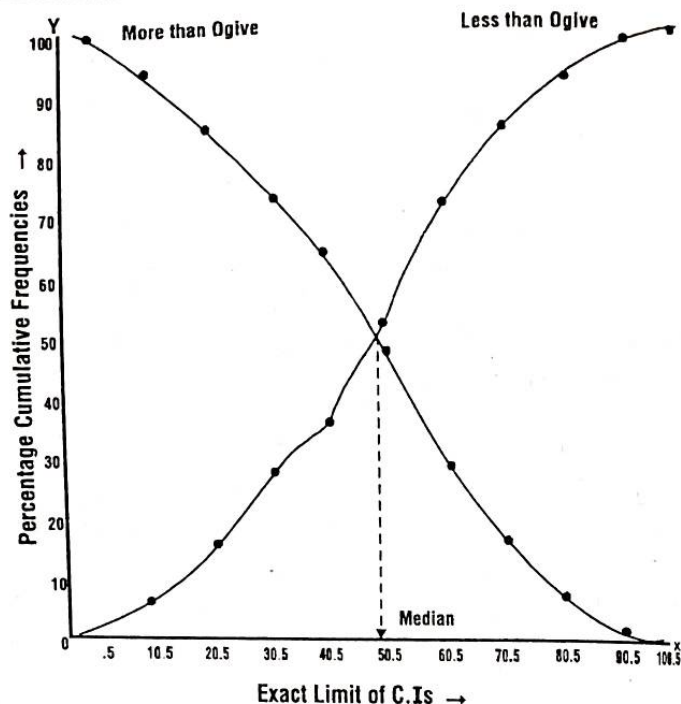
Above the exact upper limit of each class interval, the point representing the 'Less than' percentage cumulative frequency corresponding to the class interval is to be marked. If the points thus marked are joined smoothly by free hand, it results in "Less than Ogive". Using this, the values of quartiles, percentiles and percentile ranks could be found.

If, above the exact lower limit of each class interval, the point representing the 'More than' percentage cumulative frequency corresponding to the class interval is marked, then by joining these point smoothly by free hand we get the 'More than Ogive' Curve.

Generally Ogive Curves are in 'S' shape. The X - coordinate of the point of intersection of the two kinds of ogives, gives the value of the Median.

Ogive Curves

Exact limits of class Intervals	Frequency 'f'	Less than Cumulative Frequency	% Less than Cumulative Frequency	More than Cumulative Frequency	% More than Cumulative Frequency
.5-10.5	3	3	6	50	100
10.5-20.5	5	8	16	47	94
20.5-30.5	6	14	28	42	84
30.5-40.5	4	18	36	36	72
40.5-50.5	6	24	48	32	64
50.5-60.5	12	36	72	26	52
60.5-70.5	6	42	84	14	28
70.5-80.5	4	46	92	8	16
80.5-90.5	3	49	98	4	8
90.5-100.5	1	50	100	1	2
	N = 50				



Ogive Curves are useful in finding the percentiles and percentile ranks.

Conclusion

In this unit, the drawbacks of the present assessment system, assessment for better learning, assessment for confidence building in students, and assessment for creative learners, concept of 'Reflective journal', designing a good reflective journal, advantages and disadvantages of reflective journal, Concept of 'Student learning Portfolio' and using it to assess student learning, its advantages and

disadvantages, Statistical techniques for interpreting and reporting data particularly the measures of Central tendency (Mean, Median and mode), the method of computing mean, median and mode for ungrouped data (raw data) as well as for grouped data (both for discrete and continuous series), advantages and disadvantages of each of the three measures of central tendency, Measures of dispersion (Range, Mean Deviation, Standard Deviation and Quartile Deviation), special characteristics of each one of these four measures of dispersion, method of computing each one of these dispersional measures, both for the ungrouped data and grouped data (for discrete series as well as continuous series), their merits and demerits etc. Concept of 'correlation' and its types, methods of computing the correlation coefficient (both for Karl Pearson's Product Moment Correlation and Spearman's rank order correlation), Diagrams and Graphs used in statistics (Pie-diagram, Bar diagram, and Histogram, Frequency Polygon, cumulative frequency curves and Ogive curves) steps involved in drawing each of these etc. have been explained in detail.

Review Questions:

1. What are the present practices of assessing students "learning progress"? (C) [Ans: 5:01]
2. Discuss the present system of assessing students' learning progress at school education level and the drawbacks in it. (A) [Ans: 5:01 + 5:02 + 5:02:1 + 5:02:2]
3. Explain how assessment could be used for promoting better learning. (A) [Ans: 5:03]
4. Explain the salient features of 'Assessment for Confidence building in students'. (A) [Ans: 5:04]
5. What are the characteristics of creative students? (C) [Ans: In 5:05, only the first paragraph]
6. Explain the assessment for creative learners. (A) [Ans: 5:05]
7. Explain briefly the assessment for confidence building in students and the assessment for creative learners. (A) [Ans : In 5:04, except the paragraphs 2,3 and 4 + In 5:05 the first two paragraphs only]
8. What do you mean by 'reflective journal' and explain with an example how it is prepared. (A) [Ans: 5:06:1 + 5:06:2 + 5:06:3]

9. What is meant by 'reflective journal'? Explain with an example the method of designing a good reflective journal and the rubrics for evaluating it. (A) [Ans: 5:06:1 + 5:06:4 + 5:06:5]
10. Explain the concept of 'reflective journal', its advantages and disadvantages (A) [Ans: 5:06:1 + 5:06:6 + 5:06:7]
11. What do you mean by 'Student learning Portfolio'? How can it be used to assess student learning? (A) [Ans: 5:07 + 5:07:1 + 5:07:2]
12. State the advantages and disadvantages of using 'student portfolio' to assess student learning. (B) [Ans: 5:07:3 + 5:07:4]
13. What is meant by 'Measure of Central Tendency'? (C) [Ans : In 5:09, except the first paragraph]
14. What is meant by 'Arithmetic Mean'? Compute the arithmetic mean for the following data. (A)

Scores	: 20-30	30-40	40-50	50-60	60-70
Frequencies:	5	6	12	3	4

 [Ans : 5:09:1 + The arithmetic mean for the given data is 43]

15. Mention the merits and demerits of Arithmetic mean. (C) [Ans : 5:09:1:03 + 5:09:1:04]

16. What do you mean by 'Median'? Find the median for the following scores (A)

40, 30, 20, 50, 10

[Ans : 5:09:2 + Median for the given scores is 30]

17. Compute Mean and Median for the following data (A)

Scores : 30 40 50 60 75

Frequencies: 6 7 10 5 2

$$[\text{Ans : A.M} = \frac{\sum fx}{N} = \frac{1410}{30} = 47;$$

$$\text{Median} = \text{Value of } \left[\frac{N+1}{2} \right]^{\text{th}} \text{ item}$$

$$= \text{Value of } \left[\frac{30+1}{2} \right]^{\text{th}} \text{ item} = 50$$

18. State merits and demerits of 'Median' (B)

[Ans : 5:09:2:04 + 5:09:2:05]

19. What do you mean by 'Mode'? Compute the mode for the following data :

11, 21, 19, 17, 45, 19, 20, 16, 14, 23 (C)

[Ans : 5:09:3 + For the given data mode = 19 as it alone occurs the maximum number of times]

20. Compute the Mean, Median and Mode for the following data. (A)

Class Intervals : 10-20 20-30 30-40 40-50 50-60

Frequency : 5 7 12 4 2

[Ans : Mean=30.33; Median=32.5; Mode=33.64]

21. Explain the concept of 'Measures of Dispersion' and its need. Mention its important types. (B)

[Ans : 5:10:1 + 5:10:2]

22. Explain the concept of 'Mean Deviation', 'Standard Deviation' and 'Quartile Deviation'. (B)

[Ans : 5:10:4 + 5:10:5 + 5:10:6]

23. Find the Mean Deviation, Standard Deviation and Quartile Deviation for the following frequency distribution given. (A)

Class Intervals : 30-40 40-50 50-60 60-70 70-80 80-90

Frequency : 4 6 5 8 5 2

[Ans : M.D. = 11.66; S.D. 14.66; Q.D. = 11.78]

24. Write a note on each of the following :

i) Bar Diagram (B) [Ans : 5:12:1:01]

ii) Pie Diagram (B) [Ans : 5:12:1:02]

iii) Histogram (B) [Ans : 5:12:2:01]

iv) Frequency Polygon (B) [Ans : 5:12:2:02]

v) Cumulative Frequency Curve (B) [Ans : 5:12:2:03]

25. What is a 'Ogive Curve'? Explain with an illustration the method of drawing a Ogive curve and mention its uses. (A) [Ans : 5:12:2:04]
26. Explain the concept of 'Correlation between variables' and state its types. (B) [Ans : 5:11:1 + 5:11:2 + 5:11:2:01]
27. Compute Karl Pearson's Product moment correlation coefficient for the following data. (A)
- | | | | | | | |
|-----|----|----|----|----|----|----|
| X : | 19 | 23 | 27 | 30 | 26 | 24 |
| Y : | 21 | 18 | 28 | 30 | 27 | 23 |
- [Ans : $r = 0.84$]
28. Calculate the rank order correlation coefficient for the data given in question number 20. (A)
- [Ans : $\rho = 0.838$]

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ASSESSMENT FOR LEARNING

(Model Question Papers based on the revised syllabus of 2021 with answers cited in the book)

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